



A comprehensive evaluation framework from design to operation: Enhancing the regulation performance of variable-speed pumped storage plant in extreme pumping conditions

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ABSTRACT

Compared to fixed-speed pumped-storage units (FSUs), variable-speed pumped-storage units (VSUs) provide superior power adjustability in pumping mode. However, poor hydraulic stability poses a challenge to the widespread application of VSUs. Unlike previous research that only examines single aspects of VSU performance, this paper presents a comprehensive three-dimensional evaluation framework that quantifies and improves VSU resistance to hydraulic disturbances in pumping mode. This framework integrates previously disconnected subsystems including control strategies, unit combinations, and structural arrangements, providing a systematic approach to comprehensively evaluate VSU hydraulic performance. Through this research, several original insights are obtained: In control strategies, the impact of initial operating speed and control parameters on stability during pump power-off is studied. Contrary to conventional understanding, results indicate that increasing K_p helps maintain the stability of input power and speed, whereas increasing K_i is detrimental to input power stability but beneficial for speed stability. In evaluating unit combinations, differences in regulation stability between VSU + FSU and FSU + FSU schemes are assessed, revealing that while VSUs maintain shorter durations of power exceedance, they experience more pronounced speed fluctuations than FSUs. Within the safety range, a larger guide vane closing rate helps mitigate these fluctuations. Regarding structural arrangements, the study examines the effects of shared diversion pipes, shared draft pipes, and arrangements with both diversion and draft pipes shared on hydraulic disturbances. In the case study, the shared draft pipe arrangement effectively enhances the unit's resistance to hydraulic disturbances, improving speed stability by at least 63% and power stability by at least 48.7% compared to other schemes. This innovative framework illustrates the relationship between intrinsic regulation characteristics and external parameters, providing valuable references for the design and operation.

1. Introduction

1.1. Problem

Pumped storage units play an important role in the regulation and stability of grid frequency and power due to their characteristics such as fast start-up, rapid shut-down, and flexible regulation [1,2] Based on the

excitation device and rotor structure, these units are primarily classified into two types: fixed speed unit (FSU) and variable speed unit (VSU) [3]. FSU has been the preferred choice for the construction of modern pumped-storage power plant (PSPP) [4]. However, FSU regulates the output power by changing the guide vanes, a process that is relatively slow and may not suffice for the rapid regulation demands of intermittent energy sources such as wind and solar [5]. In contrast, VSU is

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increasingly valued in the power market due to its wide operating range, swift power response, flexible operational regulation, and optimal efficiency in both generating and pumping modes [6].

To save engineering investment, the project usually adopts a single-pipe, dual-unit arrangement, where the two units share a single-pressure water pipeline [7]. However, this arrangement often produces hydraulic disturbances, affecting the stability of the system. When a unit on one side undergoes power adjustment, it can disturb the flow and pressure of the normally operating unit on the other side due to hydraulic conduction between utility pipes, impacting power stability [8]. VSU achieves maximum operating efficiency by adjusting its rotational speed [9], resulting in parameters such as guide vane opening (GVO) and flows different from those of FSU. Consequently, these differences lead to more complex hydraulic disturbances than FSU. In a single-pipe, dual-unit arrangement, the stability of VSU may deteriorate due to factors such as differences in rotational speed, control parameters, and efficiency [10]. Current research on hydraulic disturbances focuses on the generating conditions of both the FSU and VSU [11–13]. However, considering the adjustability of the pumping input power in VSU, it's particularly important to study hydraulic disturbances under pumping modes. This helps the PSPP to stabilize the peak shaving and valley filling, which is essential for the stable operation of the grid. Based on this, the following questions deserve to be explored in depth:

1. What are the differences between the hydraulic disturbance effects of VSU and FSU under pumping modes?
2. How can the hydraulic disturbance effects of VSU be mitigated in pumping modes to ensure stable operation?

1.2. Literature review and gaps

Hydraulic disturbances in the single-pipe, dual-unit structure can limit the PSPP's ability to smooth grid load fluctuations. Numerous scholars have investigated various hydraulic disturbances to enhance the regulation characteristics of PSPP. Ref. [14] shows that combining the power mode of FSU with the VSU's power priority control provides better hydraulic disturbance characteristics than the FSU's frequency mode. Ref. [15] presents a nonlinear model for VSPSP with multiple units, demonstrating that VSU is more effective than FSU in reducing active power fluctuations during hydraulic disturbances. Ref. [16] compares multi-unit system transient responses and summarizes parameter impacts on stability. Ref. [17] examines the dynamic pressure response during normal operation of one unit and emergency shutdown of the other, utilizing optimized governor parameters to enhance system transient stability. A multi-objective decision-making method is proposed in Ref. [18] to identify suitable control parameters that minimize pressure fluctuations under combined conditions. Ref. [19] reveals interval time as a primary cause of extremely low draft pipe pressures during continuous load rejection. Ref. [20] analyzes the impact of parameter uncertainty on the transient characteristics during continuous load rejection in hydropower systems. Ref. [21] proposes an optimal starting strategy for parallel units to minimize both speed overshoot and the speed rise time. Ref. [22] employs a 1D-3D coupled CFD approach to simulate simultaneous load rejection in two pump turbines, validating its engineering feasibility. Ref. [23] examines hydraulic and mechanical parameter impacts on parallel pumped storage unit vibration, showing that equal load variation distribution improves stability over unequal distribution.

The main difference between VSU and FSU is the replacement of the conventional synchronous motor with a doubly-fed induction generator. In exploring the advantages and evaluation of VSU, Ref. [24] quantitatively assesses the multi-timescale frequency modulation performance of VSU in wind-pumped storage systems, showing it mitigates power and frequency fluctuations. Ref. [25] analyzes the effects of various speed control instructions on the VSU and proposes a suitable VSU speed control scheme that balances speed and safety considerations. Ref. [26]

compares the flexibility advantages of VSU over FSU during specific frequency modulation events. The study by Ref. [27] shows that the day-ahead market, ancillary services, and the intraday market yield the highest total profit in electricity markets with VSU. Ref. [28] constructs a joint simulation model of wind power and pumped storage, proving that VSU exhibits better indexes including the average and standard deviation of power difference. Ref. [29] compares the power regulation performance of VSU and FSU in both power generation and energy storage modes, verifying the regulation reliability under pumped storage operation. The study in Ref. [30] uses numerical simulations to compare doubly-fed induction machines (DFIM) and synchronous motors in VSU and FSU, finding that DFIM offers significant advantages for grid support and renewable energy integration. The above studies on VSU have concentrated on the superiority of the frequency modulation performance in power generating conditions, while the resilience of VSU to disturbances in pumping mode has received less attention. It should be noted that the instability of the VSU's power absorption from the grid increases the complexity of hydraulic disturbance characteristics. Therefore, conducting an in-depth investigation into the stability of VSU and proposing measures to improve it are essential to enhance the regulation of the grid.

1.3. Contribution

This paper addresses the research gaps mentioned above by conducting the following innovative studies:

1. A pumping-generation system model of a single-pipe, dual-unit pumped storage system is constructed, which includes both VSU and FSU. This model aims to study the nonlinear characteristics of power, rotational speed, and pressure in pumping modes, providing a model basis for the comprehensive evaluation framework.
2. Based on the VSU pumping-generation system model, the evaluation framework incorporating the control strategies, the unit combinations, and the structure arrangements is established. The hydraulic disturbance characteristics of variable speed pumped-storage power plant (VSPSP) are analyzed, revealing the implicit relationships between control parameters and stability indicators such as power and rotational speed.
3. Based on the evaluation framework, optimal strategies for enhancing resistance to hydraulic disturbances are proposed from operation and design perspectives. These strategies include control parameters, a guide vane closing strategy, the effective unit combination scheme, and the optimal pressure pipe arrangement.

The structure of this paper is as follows: Section 2 presents the methodology of this paper and the proposed integrated evaluation framework. Section 3 studies the impacts of operating parameters on pump power-off hydraulic disturbances. Section 4 investigates the influence of unit combinations on the results of disturbances. Section 5 examines the effects of the pipe arrangements on pump power-off disturbances. Section 6 provides the conclusion and discussion.

2. Model

2.1. Comprehensive evaluation framework

Superior transient performance is an essential factor in the design of VSPSP and is also a guarantee of reliable operation. Factors affecting transient performance include control strategy, pump turbine type, and pipeline arrangement [31]. Among them, the control strategy determines the length of the regulation time, which affects the auxiliary service revenue of the power plant. The type of unit influences the recovery speed after disturbances, while the pipeline arrangement impacts both the propagation path of the water hammer wave and its effect on the pump turbine. Therefore, optimizing key parameters from design to

operation is crucial for improving the stability of VSPSP. This paper develops a comprehensive evaluation and optimization framework that assesses the operating factors and design elements of VSPSP from a full life cycle perspective. Additionally, the framework dynamically adjusts the operating parameters based on the evaluation results to mitigate hydraulic disturbances, thereby optimizing the pumping process.

As shown in Fig. 1, the evaluation framework consists of four key components. The modeling part serves as the foundation of the entire framework. In the unit operational performance evaluation section (as shown in the upper right corner of Fig. 1), this study thoroughly explores the response characteristics of the VSU to hydraulic disturbances at different initial rotational speeds. Additionally, it evaluates the impact of various governor parameters and guide vane closing rules on the transient response of the disturbed unit and proposes an optimal control strategy based on these findings. In the unit combination evaluation section (the lower left corner of Fig. 1), this study analyzes the transient performance differences between the disturbed FSU and VSU during pump power-off and pump start-up conditions. It examines two unit combination types: VSU + FSU and FSU + FSU, developing an optimal guide vane control strategy for each. Lastly, in the structural arrangement evaluation section (the bottom right corner of Fig. 1), this study compares three different structural arrangements to uncover the intrinsic mechanisms of the transient power and pressure response. With these insights, it identifies the most effective pipe arrangement scheme to enhance resilience against hydraulic disturbances. The following sections describe the mathematical model and principles of this framework (see Fig. 2).

2.2. Hydraulic subsystem

When the flow velocity at the end of the pressure pipe in the pipeline system changes, it causes an unsteady flow within the pipe. This leads to significant pressure fluctuations due to variations in flow velocity, commonly referred to as water hammer. During frequent shifts in

working conditions of the pump turbine, the continuous adjustment of guide vanes often triggers the water hammer phenomenon. In this paper, the method of characteristics is employed to refine the modeling of the pipeline system.

The unsteady flow within 1-D equations is characterized by the continuity and momentum equations, formulated as a partial differential Eq. (1) [32]:

$$\begin{cases} L_1 = \frac{\partial H}{\partial x} + \frac{V}{g} \frac{\partial V}{\partial x} + \frac{1}{g} \frac{\partial V}{\partial t} + \frac{fV|V|}{2gD} = 0 \\ L_2 = \frac{\partial H}{\partial t} + V \frac{\partial H}{\partial x} + \frac{a^2}{g} \frac{\partial V}{\partial x} + V \sin \alpha = 0 \end{cases} \quad (1)$$

where H denotes the piezometric head at the pipe cross section in m ; V is the flow velocity in m/s ; x represents the distance from the upstream section in m ; D is the diameter of the pipe in m ; a signifies the velocity of the water hammer wave in m/s ; f is the friction coefficient; g is the acceleration of gravity in m/s^2 .

The partial differential Eq. (1) is discretized into the difference Eq. (2) by setting the time and pipe as Δt and Δx , respectively, and integrating along the characteristic lines $C+$ and $C-$ in Fig. 3:

$$\begin{cases} C^+ : H_p = H_A - \frac{a}{gA} (Q_p - Q_A) - \frac{fQ_p|Q_p|\Delta x}{2gDA^2} = 0 \\ C^- : H_p = H_B - \frac{a}{gA} (Q_p - Q_B) + \frac{fQ_p|Q_p|\Delta x}{2gDA^2} = 0 \end{cases} \quad (2)$$

where the subscript p stands for the variable at time $t + \Delta t$, while subscripts A and B are the variables on characteristic lines $C+$ and $C-$ at time t . At time $t = 0$, the system is in a steady flow state, with known flows and water heads at each pipe section. The parameters at time $t = \Delta t$ are determined by solving the conditions at $t = 0$, followed by a subsequent solution for the conditions at $t = 2\Delta t$. Eq. (2) can be simplified as follows:

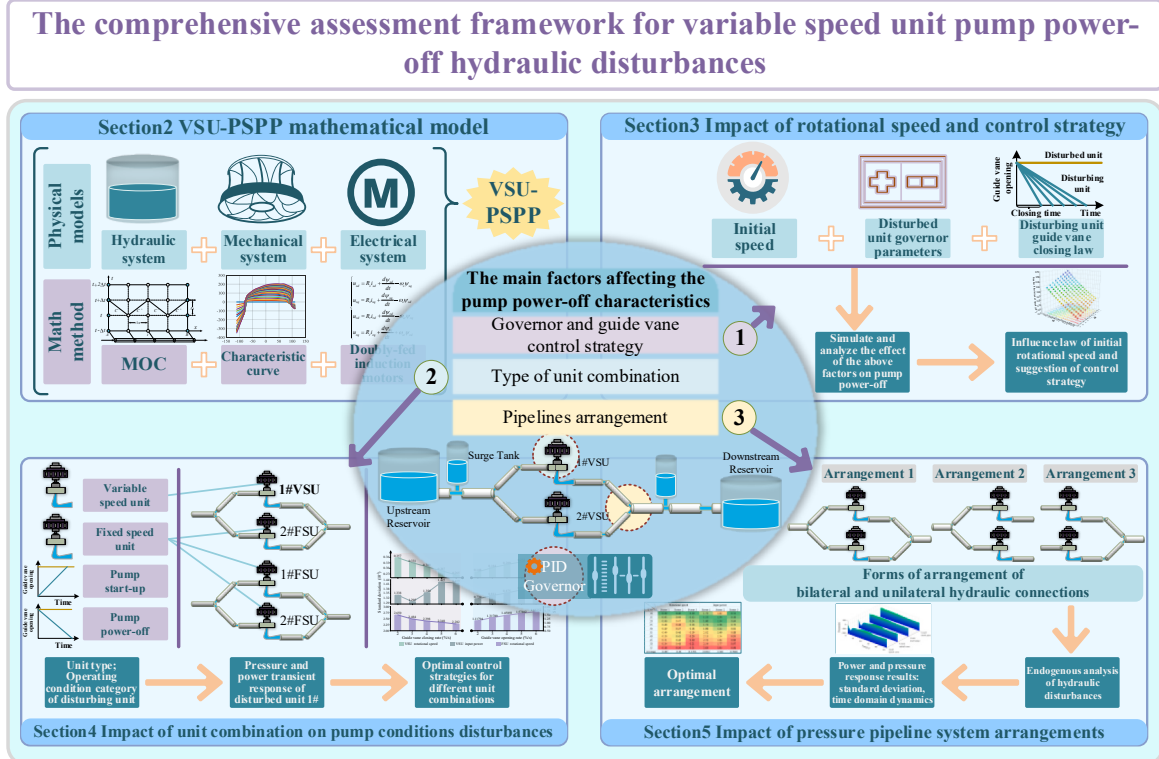


Fig. 1. Comprehensive evaluation framework.

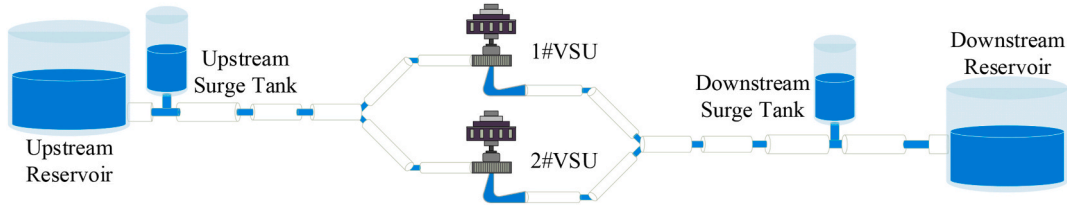


Fig. 2. Structural arrangement of pumped-storage power plants.

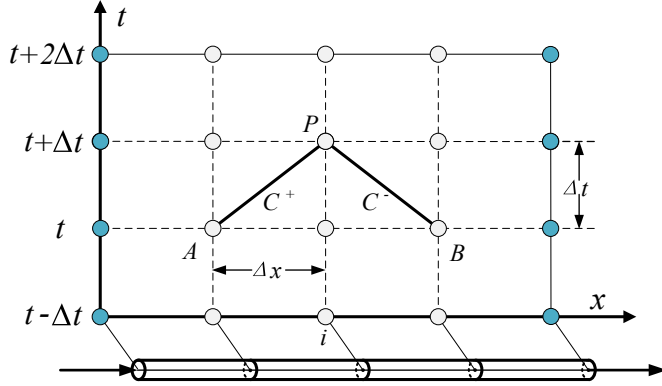


Fig. 3. The method of characteristics.

$$\begin{cases} C^+ : H_p = C_p - B_p Q_p \\ C^- : H_p = C_M - B_M Q_p \end{cases} \quad (3)$$

where C_p , B_p , C_M , and B_M are known quantities at time t , and their calculation formulas are as follows:

$$\begin{cases} C_p = H_{i-1} + BQ_{i-1} \\ B_p = B + R|Q_{i-1}| \\ C_M = H_{i+1} - BQ_{i+1} \\ C_M = B + R|Q_{i+1}| \end{cases} \quad (4)$$

where B and R are known quantities, and their expressions are given as:

$$\begin{cases} B = \frac{a}{gA} \\ R = \frac{f\Delta x}{2gDA^2} \end{cases} \quad (5)$$

In the pipeline system, Q and H are determined by Eq. (2). However, special nodes, such as upstream and downstream reservoirs, series pipes, bifurcated pipes, convergent pipes, surge tanks, and units, are subject to only one equation [32]. These nodes require additional boundary conditions to be solved (The equations can be found in Appendix A).

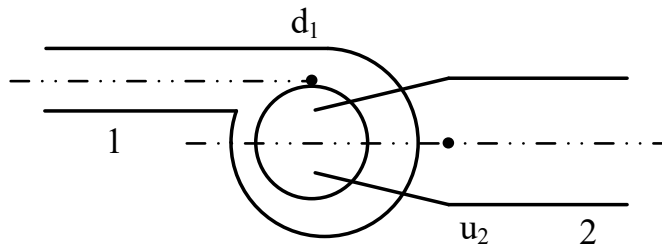


Fig. 4. The pump turbine.

2.3. Mechanical subsystems

2.3.1. Pump turbine

For Pipes 1 and 2, as illustrated in Fig. 4, the existing compatibility equations for both sides are:

$$C^+ : H_{p1} = C_{p1} - B_{p1} Q_{p1} \quad (6)$$

$$C^- : H_{p2} = C_{M2} - B_{M2} Q_{p2} \quad (7)$$

The unit node serves as the intersection for the two pipes and necessitates the introduction of two additional auxiliary equations. Given the proximity of the d_1 and u_2 sections, they may be approximated as follows:

$$Q_{p1} = Q_{p2} = Q_t \quad (8)$$

The water head acting on the unit and the flow of the unit are as follows:

$$H_t = H_{p1} - H_{p2} \quad (9)$$

$$Q_t = Q_{11} D_1^2 \sqrt{H_t} \quad (10)$$

where Q_{11} is a function of the guide vane opening and the unit rotational speed, and it can be determined from the complete characteristic curve of the unit. The simultaneous equations can be solved to obtain the parameter values at the hydraulic unit node.

The pump turbine acts as a major component of the mechanical system, utilizing flow and torque characteristic curves (Fig. 5) to calculate its transient processes [23]. However, the S characteristic region and hump region of the complete characteristic curve can cause data interpolation errors during simulation, where a single N_{11} value corresponds to multiple Q_{11} values on the same guide vane line. To ensure simulation accuracy, this paper employs the improved Suter Transformation method to process the complete characteristic curve [16]. The comprehensive characteristic curves processed and fitted using the Suter Transformation method are shown in Fig. 6, and these processed curves eliminate the multivalued problem.

The improved Suter Transformation method:

$$\begin{cases} W_h(x, y) = \frac{h}{a^2 + q^2 + C_h h} (y + C_y)^2 \\ W_m(x, y) = \left(\frac{m + k_1 h}{a^2 + q^2 + C_h h} \right) (y + C_y)^2 \end{cases} \quad (11)$$

$$\begin{cases} x = \arctan \frac{q + k_2 \sqrt{h}}{a} & a > 0 \\ x = \pi + \arctan \frac{q + k_2 \sqrt{h}}{a} & a < 0 \end{cases} \quad (12)$$

in Eqs. (11) and (12), x is the relative flow angle; a , q , h , m , y are the values of unit rotational speed, flow, water head, torque, and opening, respectively; k_1 , k_2 , C_y , C_h is the basic parameter.

2.3.2. Unit iteration process

The energy conversion process between the pump turbine and the pressure pipeline system is a complex, nonlinear problem, and a change

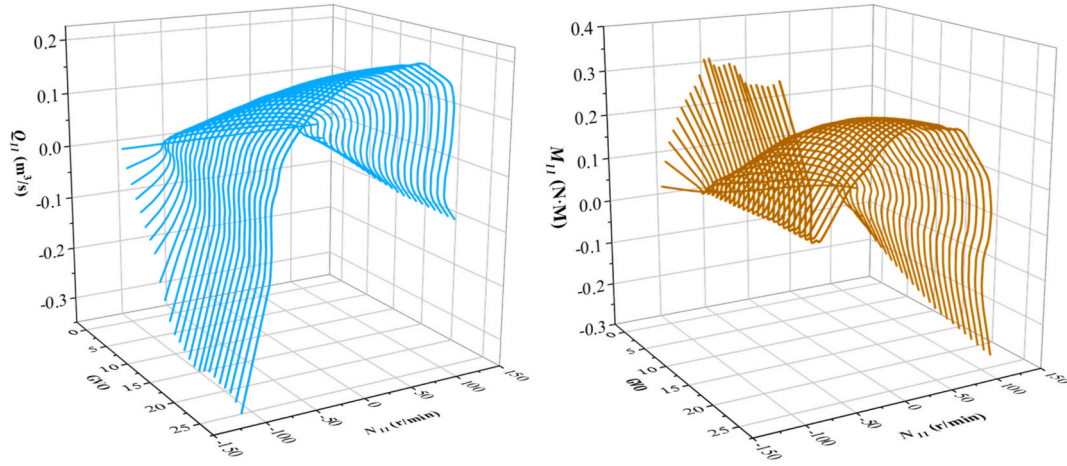


Fig. 5. Flow and torque characteristic curves of the pump turbine.

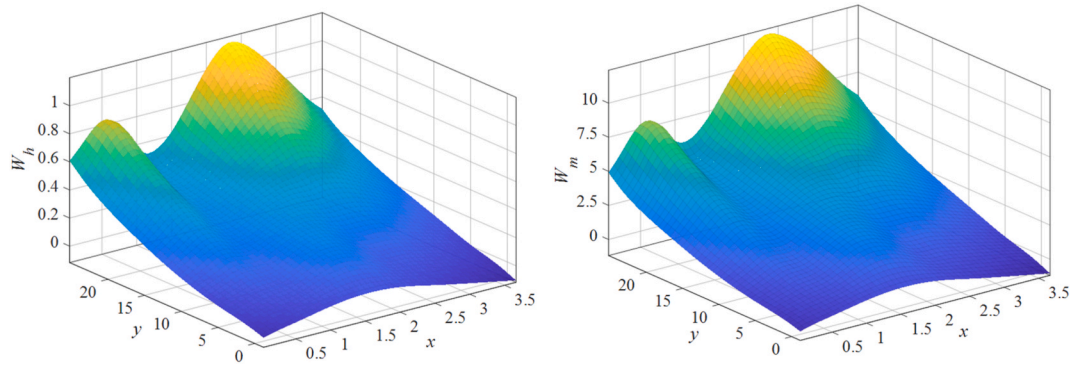


Fig. 6. Complete characteristic curves of improved suter transformation method.

in any parameter can affect the operation of the entire system. To accurately simulate the data exchange process between the pump turbine and the pressure pipeline system, this paper employs a rotational speed iterative loop nested within a flow iterative loop for convergence calculations [33]. The iterative process within a single time step is depicted in Fig. 7.

2.4. Electrical subsystem

The VSU in this paper utilizes a doubly-fed induction generator for variable speed and constant frequency operation through control of the rotor-side excitation. The three-phase stationary coordinate axes of the model are transformed into a two-phase rotating d - q coordinate frame, applying the principle of equal amplitude [7].

(1) Voltage equation for doubly-fed asynchronous generator in d - q coordinate frame

$$\begin{cases} u_{sd} = R_s i_{sd} + \frac{d\psi_{sd}}{dt} - \omega_s \psi_{sq} \\ u_{sq} = R_s i_{sq} + \frac{d\psi_{sq}}{dt} + \omega_s \psi_{sd} \\ u_{rd} = R_r i_{rd} + \frac{d\psi_{rd}}{dt} - \omega_{sr} \psi_{rq} \\ u_{rq} = R_r i_{rq} + \frac{d\psi_{rq}}{dt} + \omega_{sr} \psi_{rd} \end{cases} \quad (13)$$

Where the subscripts s , r , d , and q correspond to the stator, rotor, d -axis, and q -axis, respectively; i is the current; ψ is the magnetic flux linkage; ω_s is the stator frequency.

(2) Magnetic flux linkage equation

$$\begin{cases} \psi_{sd} = L_s i_{sd} + L_m i_{rd} \\ \psi_{sq} = L_s i_{sq} + L_m i_{rq} \\ \psi_{rd} = L_r i_{rd} + L_m i_{sd} \\ \psi_{rq} = L_r i_{rq} + L_m i_{sq} \end{cases} \quad (14)$$

where L_m is the magnetizing inductance; L_s is the stator inductance; and L_r is the rotor inductance.

(3) Active and reactive power and electromagnetic torque equations

$$\begin{cases} P_s = \frac{3}{2} (u_{sd} i_{sd} + u_{sq} i_{sq}) \\ Q_s = \frac{3}{2} (u_{sq} i_{sd} - u_{sd} i_{sq}) \\ T_e = \frac{3}{2} n_p L_m (i_{sq} i_{rd} - i_{rq} i_{sd}) \end{cases} \quad (15)$$

where n_p is the number of magnetic poles.

(4) Generator speed

$$J \frac{d\omega}{dt} = T_t - T_e \quad (16)$$

where J is the moment of inertia of the rotating part of the unit; ω is the rotational angular velocity of the unit in rad/s ; $\omega = \pi n/30$, n is the rotational speed of the unit in r/min ; T_t is the mechanical torque in $\text{N} \cdot \text{m}$.

(5) Rotational speed and power control systems

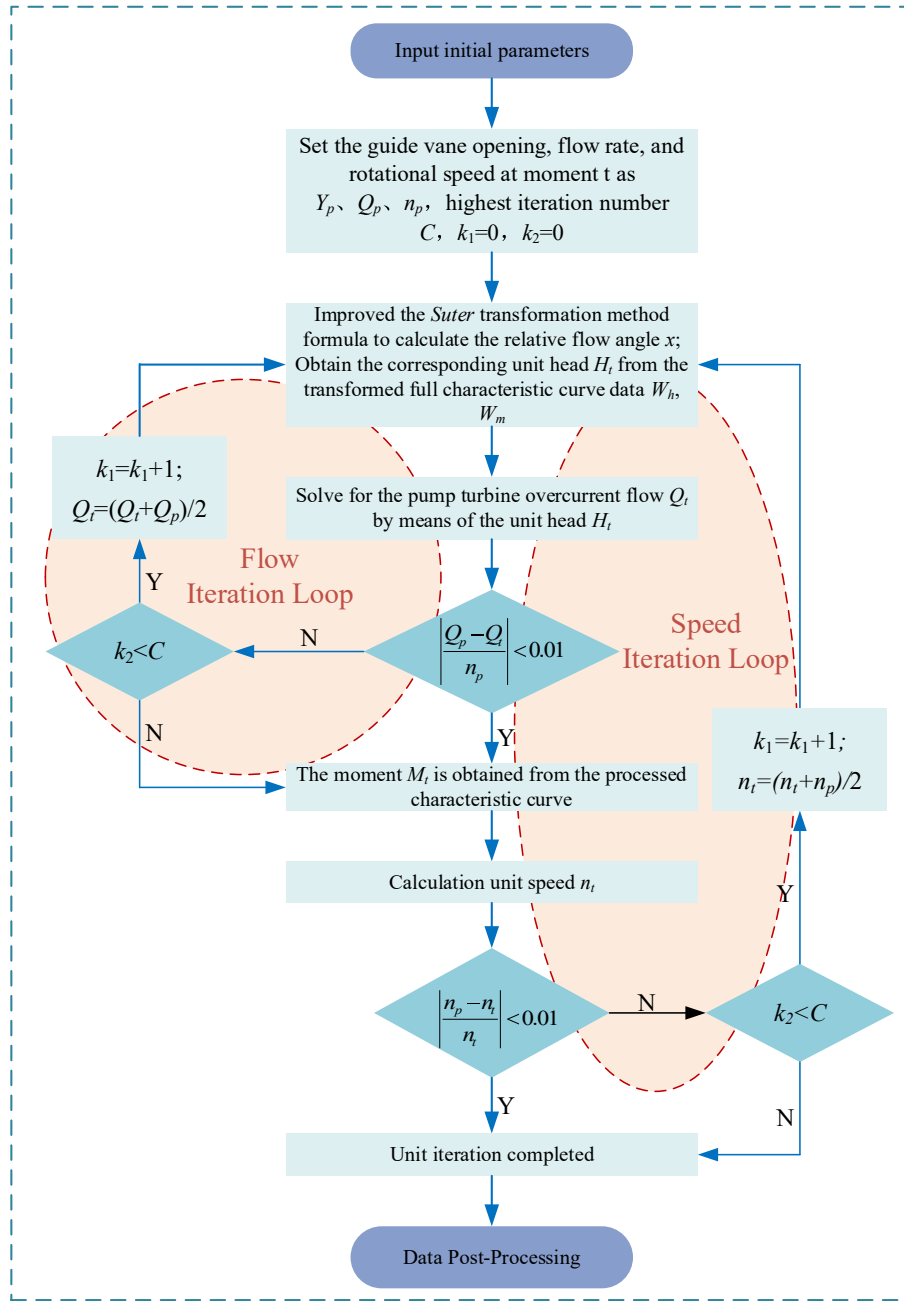


Fig. 7. Refined model iterative flows for pumped storage units.

In pumping mode, the guide vane opening (GVO) has a limited regulatory effect on unit power. Therefore, the GVO is fixed at the maximum efficiency point in this study. This is because the active power command and static head operate on longer time scales (minutes to hours), and within the electromechanical time scale, the optimal guide vane opening can be considered to remain constant. Under this condition, the entire control system is directly executed by the excitation regulation system.

Depending on the direct control target of the excitation regulation system, VSU in pumping mode can adopt two control strategies: (1) Fast power control (power priority control): The speed deviation is processed by a PI controller to fine-tune the power setpoint, and the power is directly controlled by the excitation regulation system; (2) Fast speed control (speed priority control): The power deviation is processed by a PI controller to fine-tune the speed setpoint, and the speed is directly controlled by the excitation system. Schematic diagrams of these two

control strategies are shown in Fig. 8.

In modern power systems with high penetration of renewable energy, VSUs should primarily meet the requirements of rapid response in automatic generation control to better mitigate disturbances from renewable energy generation. Therefore, power priority control is typically the preferred strategy. The dynamic model of the VSU established in this paper adopts the widely recommended power priority control strategy.

The conclusions of this paper are applicable to the power priority control scenario. In all regulation simulations presented in this paper, the VSU serves as the disturbed unit. Due to the short time scale of the disturbance (on the order of seconds), the GVO of the disturbed unit is maintained at the optimal efficiency point, while the rotational speed freely responds to hydraulic disturbances within the allowable range. The electromagnetic power is maintained stable by the converter through regulation of the electromagnetic torque. However, the

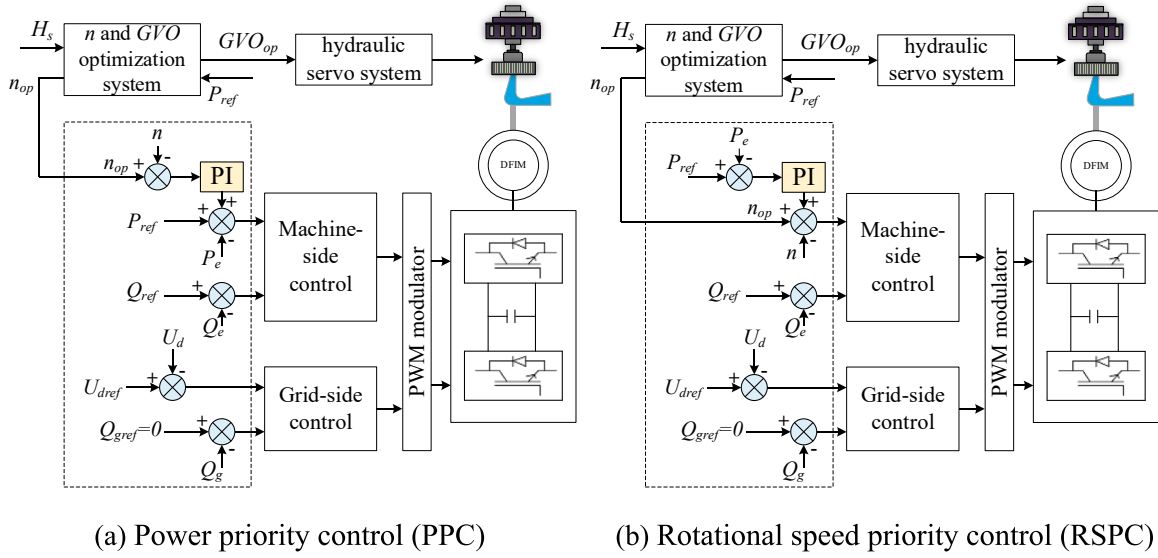


Fig. 8. Schematic diagram of VSU pumping mode control strategy.

hydraulic parameters of the disturbed unit, including rotational speed, pressure, and mechanical torque, experience fluctuations due to water hammer effects, leading to corresponding changes in mechanical power. We have conducted comparative experiments examining hydraulic disturbances under both power priority control and speed priority control with different PI parameters. The results are presented in Appendix E.

2.5. Efficiency and input power characteristics of VSU in the pump region

In the operation of VSU, the optimal rotational speed and Guide Vane Opening are determined through rotational speed optimization for a given head and power [34]. The equal efficiency surfaces and equal input power surfaces of the pumped storage units used in this study are shown in Fig. 9.

Fig. 9(a) shows the equal efficiency surface of the pumped storage units, where warm colors represent high-efficiency operating regions, and cool colors represent low-efficiency regions. The green line represents the steady-state operating curve of the FSU at the rated head, while the blue point represents the rated operating point of the FSU in pump mode. It can be seen that at small openings, the larger the guide vane opening, the higher the pumping efficiency (The figure inspired by Ref. [35]). Fig. 9(b) shows the equal input power surface, where warm

colors indicate higher input power for the units, and cool colors indicate lower input power. Combined with Fig. 9(a), it is evident that a lower unit speed under the same input power and head can improve operating efficiency, but the speed cannot be reduced indefinitely. Otherwise, it will go beyond the characteristic curve.

Due to the variable rotational speed characteristics of the VSU, the initial rotational speed is crucial for the stable operation of the unit. In this section, seven representative steady-state operating points on the same input power line are pre-selected, as indicated by the black circles in the figure. On the same input power line, the larger the guide vane opening, the closer it is to the high-efficiency region. However, increasing the guide vane opening also increases the flow, which can affect the unit's regulatory quality during transients. Therefore, Section 3.2 investigates the impact of the initial rotational speed on hydraulic disturbances under pump conditions.

To ensure the correctness of the model, this paper selects a PSPP in China that is already operational for simulation and experimental comparison. The results are presented in Appendix B. Additionally, all simulations in this paper are based on MATLAB 2019a.

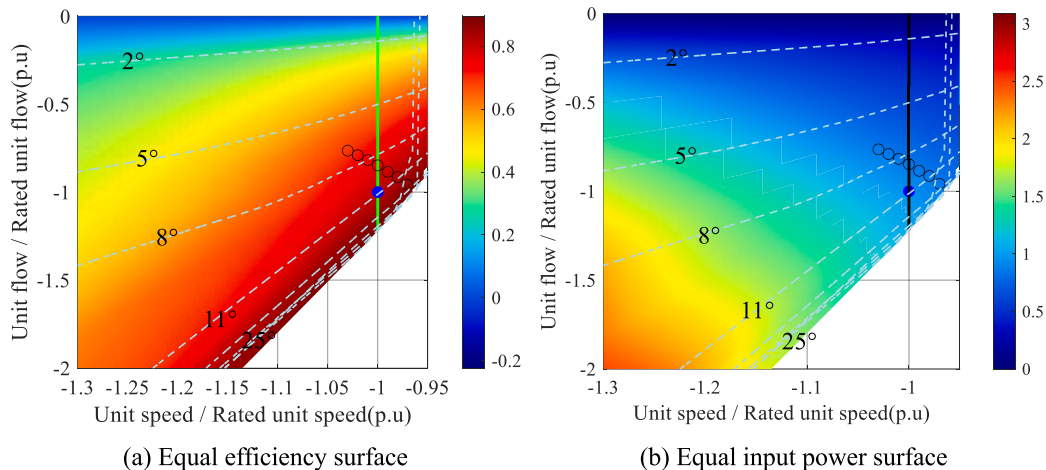


Fig. 9. Efficiency surface and input power surface for the pump mode region.

3. Impact of operating parameters - rotational speed and control strategy on VSU hydraulic disturbances

The transient response of an operating VSU after hydraulic disturbances depends on three key factors. Firstly, the rotational speed variability determines the steady-state operating conditions of VSU when experiencing hydraulic disturbances. Secondly, the guide vane closing rules determine the intensity of these disturbances. Finally, the governor is an important regulating element and determines the resilience of the disturbed unit. This section analyses the effects of initial operating rotational speed, governor parameters and guide vane closing rules on the transient performance of the VSPSP.

3.1. Initial operating point and pump power-off pressure response characteristics

To reveal the hydraulic disturbance characteristics of the single-pipe, dual-unit pumping condition, this subsection simulates the transient process of VSU during pump power-off under rated power and specified water head conditions. The pressure responses in the time-domain for the disturbing and disturbed units are fully analyzed, with the corresponding results shown in Figs. 10 and 11. It is worth noting that VSU has speed adjustability and requires optimal initial operating conditions, which are found through the speed optimization strategy. The optimal results are an initial operating speed of -420.46 r/min, a guide vane opening of 83%, and a pump flow of 66.86 m³/s. These parameters ensure that VSU operates at rated input power with maximum pumping efficiency, representing the optimum steady-state operating conditions.

To define the characteristic phases of the pump power-off process, Fig. 10(a) shows the pump power-off process phases for the disturbing unit. Based on the time of flow and rotational speed zero points, the entire process is categorized into four distinct phases: normal speed with forward flow (NSFF), normal speed with reverse flow (NSRF), reverse speed reverse flow (RSRF), pump shutdown and runaway (PSR). These phases correspond to different working zones on the characteristic curve: pump working zone, pump coasting zone, turbine working zone, and reverse pumping zone. Pumping mode with forward flow and pumping mode with reverse flow phases are of shorter duration, while the generating mode with reverse flow and pump shutdown and runaway phases are longer. Fig. 10(b) shows the speed-flow trajectories of the disturbing unit under different guide vane closing times. As illustrated, not all pump shutdowns will experience all four stages completely. The final stage to which the unit returns during shutdown depends on the guide vane closing time. We categorize the guide vane closing times into three types: rapid closure, uniform closure, and slow closure.

- Rapid closure (10, 15, 20 s): The operating point returns from the pump coasting zone to shutdown status, ultimately experiencing the NSRF stage.
- Uniform closure (25, 30, 35, 40 s): The operating point returns from the turbine working zone to shutdown status, ultimately experiencing the RSRF stage.
- Slow closure (45, 50, 55 s): The operating point returns from the reverse pumping zone to shutdown status, ultimately experiencing the PSR stage.

Fig. 11 depicts the time domain response of the spiral case pressure for the disturbing and disturbed units. From the figure, it is evident that while the response trend of the disturbed unit is similar to that of the disturbing unit, the minimum pressure is lower. In addition, there is a time deviation between the two pressure minima. These phenomena occur because the water hammer wave takes time to propagate through the pipe, and the friction between the water and pipe wall gradually weakens the pressure fluctuations at the same time.

Fig. 12 displays the draft pipe pressure time domain response of the disturbing and disturbed units. In the 2 s before the power-off, the draft tube pressure trend of the disturbed unit is opposite to that of the disturbing unit, but consistent with the spiral case pressure trend. The reason for this phenomenon is:

Following pump power-off, the water hammer wave impacts the disturbed unit through the diversion and draft pipes, leading to changes in water head. As the diversion point is more significantly affected, the trend of water head change is similar to that of spiral case pressure, but with a weaker magnitude. Since unit water head is defined as the difference between the spiral case pressure and the draft pipe pressure, this effect of reduced magnitude is reflected in the draft pipe pressure. Consequently, the change in draft pipe pressure initially follows the same trend as the change in spiral case pressure. This phenomenon of draft pipe pressure following spiral case pressure is commonly observed during various working condition transitions and disturbances, including the load rejection process, pump power-off, and other scenes.

3.2. Impact of initial operating rotational speed

The variable speed characteristic of VSU is crucial, directly influencing the hydraulic characteristics and power control performance of the disturbed VSU. This subsection explores the effect of initial VSU speed on hydraulic disturbance characteristics during pump power-off. The disturbing unit is set for 90% of the input power at the rated head. After optimization of the rotational speed and guide vane, the initial rotational speed is -412.18 r/min, the initial guide vane opening is 62%, and the guide vane closes over 30 s. The disturbed unit is set to

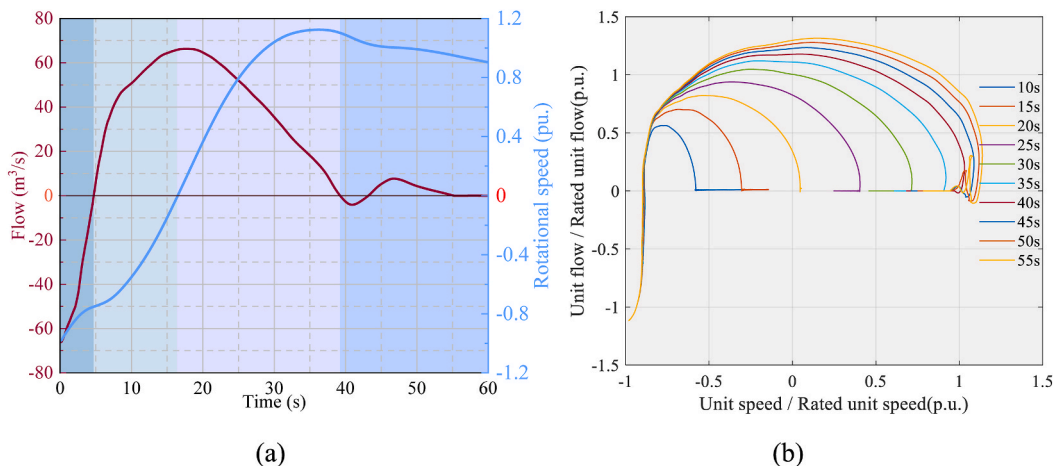


Fig. 10. Schematic of the phases of disturbing unit pump power-off condition.

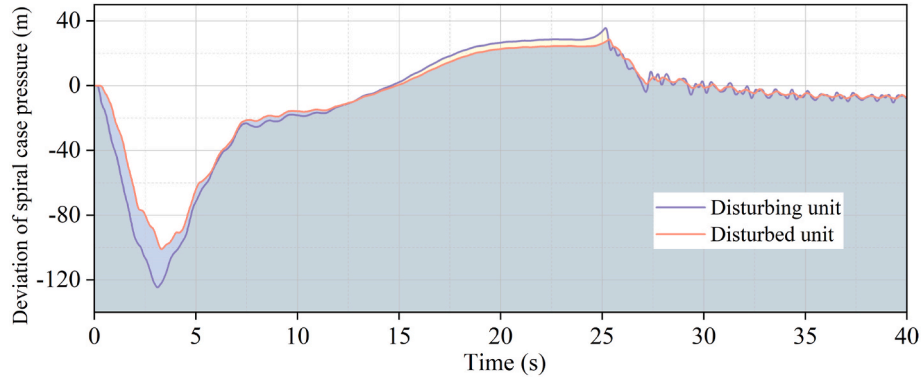


Fig. 11. Time domain response of spiral case pressure.

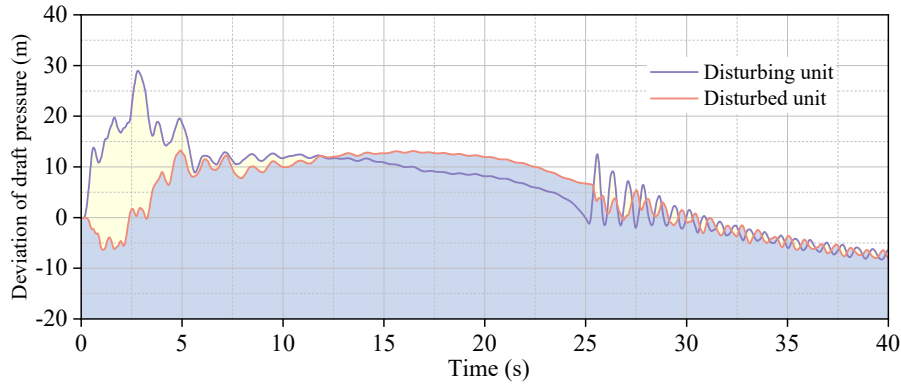


Fig. 12. Time domain response of draft pipe pressure.

operate at 90% of the input power at the rated head, with the rotational speed varying from -0.97 p.u. to -1.03 p.u. across 7 distinct values. In this set of conditions, the disturbed units have the same input power, water head, and disturbance level, despite different initial operating rotational speeds.

It should be noted that doubly-fed variable speed units should typically avoid synchronous operation (-1 p.u.). Although this study includes -1.0 p.u. as one of the test cases to ensure comprehensive

analysis, synchronous operation is generally not recommended in practical applications. This is because synchronous operation may reduce the control flexibility of the rotor-side converter and potentially compromise system stability. To maximize the efficiency advantages of doubly-fed variable speed units, guide vane and rotational speed optimization should be implemented rather than operating at synchronous speed.

Based on the setup parameters of the disturbing and disturbed units,

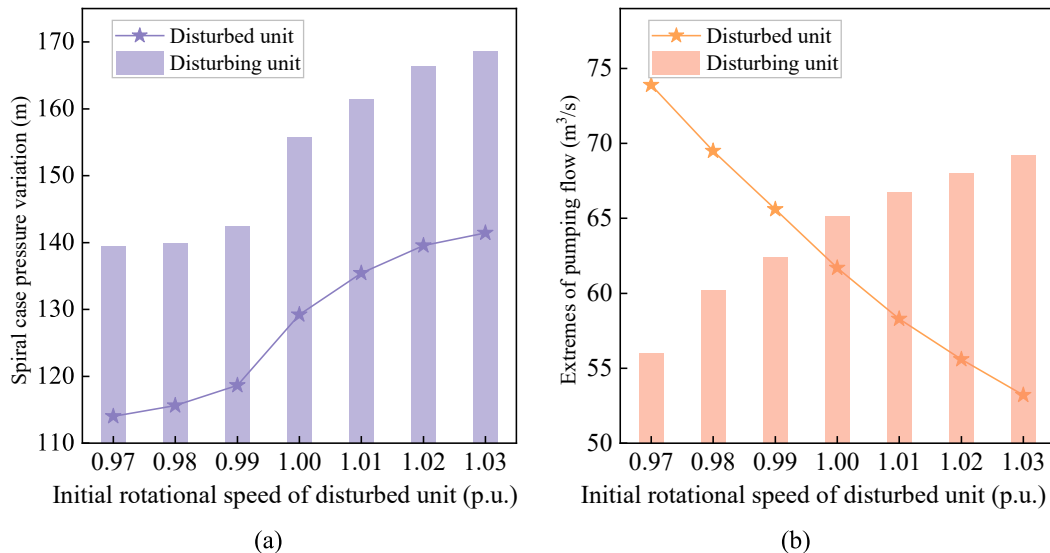


Fig. 13. The spiral case pressure and flow variations at different initial rotational speeds.

a numerical simulation of the hydraulic disturbance transition process is conducted at various rotational speeds. The resulting variations in spiral case pressure and unit flow for these units are depicted in Fig. 13.

Fig. 13(a) shows the spiral case pressure variations between the disturbed and disturbing units at different initial rotational speeds. The results indicate that the variations in spiral case pressure for both the disturbed and disturbing units are positively correlated with the initial operating rotational speed. Fig. 13(b) compares the extremes of pumping flow for the disturbed unit and backflow for the disturbing unit. As shown in the figure, the pumping flow at a speed of 1.03 p.u. is $20.7 \text{ m}^3/\text{s}$ lower than that at 0.97 p.u. speed for the disturbed unit. Additionally, there is a negative correlation between the extreme value of the pumping flow and the unit initial rotational speed, indicating that a higher speed is associated with a better flow response. Conversely, the backflow in the disturbing unit is positively correlated with its initial rotational speed. This correlation can be attributed to the reverse disturbances within the unit, leading to a distinct trend in flow.

In summary, a higher initial operating rotational speed of the disturbed unit tends to result in poorer pressure characteristics but improved flow characteristics. Furthermore, when a unit is disturbed, it

can impact the dynamic characteristics of the disturbing unit, with the impact diminishing as the initial rotational speed decreases.

3.3. Results and improvement of control strategies

During hydraulic disturbances, in addition to the initial conditions, the control strategy significantly influences the regulation performance of the disturbed unit. This subsection analyzes the effects of guide vane closing time and governor parameters on the hydraulic disturbances during pump power-off in VSU.

3.3.1. Guide vane closing time on disturbing unit

The analysis from Section 3.1 indicates that the unit experiencing pump power-off will go through various stages to achieve shutdown, which has different effects on the pressure changes of the disturbed unit. To examine the impact of the disturbing unit entering different stages, the VSU initially carried the rated input power with one unit performing the power-off disturbance while the other operated normally as the disturbed unit. The guide vane closing time of the pump power-off unit was set from 10 to 55 s, increasing in 5 s intervals to simulate the effect

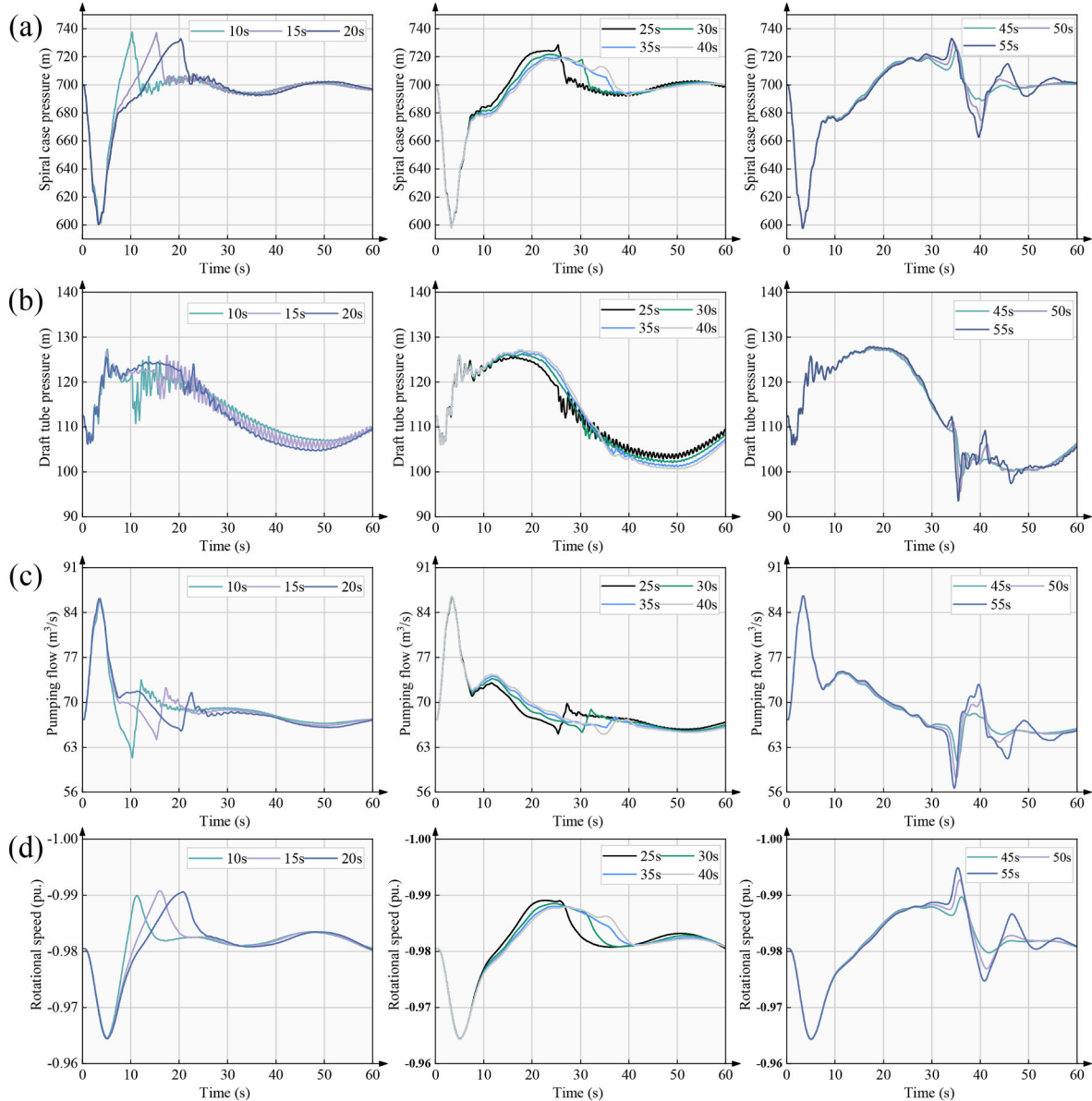


Fig. 14. Transient indicator response at different guide vane closing times.

of disturbances at different closing rates. The obtained time-domain dynamic response characteristics of the disturbed unit in terms of pressure, flow, and rotational speed through simulation are shown in Fig. 14(a) to (d), indicating that:

- (1) Fig. 14(a) and (b) demonstrate the effect of different guide vane closing times on pressure fluctuations. During the first 6 s, all pressure curves overlap, indicating that the disturbing unit operates in NSFF and NSRF. Subsequently, the curves begin to disperse as the closing time increases. In particular, significant increases are observed in the extreme values of the spiral case pressure and fluctuations occur in the draft pipe pressure at closing times of 10, 15, and 20 s. From 25 to 40 s, the pressures in the spiral case and draft pipe become flat and stabilize after the initial peaks and valleys. However, when the closing time extends to 45, 50, and 55 s, fluctuations in both the spiral case and draft pipe pressures increase. This is because the disturbing unit enters the PSR stage, and the pressure fluctuations caused by the S region affect the disturbed unit. This results in the unique phenomenon where the operating point of the disturbed unit is located in the pump characteristic zone, but the pressure variations resemble those of the S region.
- (2) Fig. 14(c) shows the time-domain characteristics of the pumping flow of the disturbed unit. Initially, the pumping flow surges to a peak, then decreases, and finally stabilizes after some fluctuations. When the disturbing unit starts transitioning operating modes, its guide vanes begin to close, causing the pump flow to drop. However, since the flow supply from the downstream reservoir remains constant, the loss of pumping capacity from the disturbing unit transfers flow to the normal operating disturbed unit, leading to a temporary increase in flow. As the disturbing unit transitions into the NSRF phase, its flow starts to reverse, and the inflow to the disturbed unit gradually decreases. The pumping flow then begins to drop, and the time-domain flow curves for different closing times start to diverge. During rapid closure, the flow stabilizes gradually after fluctuations between 7 and 25 s. For moderate closure, fluctuations occur between 25 and 37 s before stabilizing. In the case of slow closure, fluctuations are more intense, occurring between 34 and 50 s before stabilizing. This is because the corresponding disturbing unit enters the PSR phase, resulting in more significant flow fluctuations for the disturbed unit.
- (3) Fig. 14(d) shows that at the start of the disturbance, the actual rotational speed of the disturbed unit increases, peaks, and subsequently decreases to stabilize after fluctuations. When the disturbing unit switches working conditions, both mechanical and electromagnetic torques of disturbed unit increase. However, the increase in electromagnetic torque lags behind the mechanical torque, leading to a increase in rotational speed. Subsequently, the mechanical torque of the unit is adjusted with the resistance torque to promote a gradual stabilization of the rotational speed. The rotational speed fluctuation remains within a range of 2.35% to 3.05% across different guide vane closing times.

To investigate the influence of different guide vane closing times on the regulation performance of the disturbed unit, the standard deviation of rotational speed and input power is selected as the evaluation index. Since the speed and power of the disturbed unit eventually return to their original values, the standard deviation can appropriately reflect the fluctuation state of the unit's speed and power. The calculation method for standard deviation is as follows:

$$\sigma = \sqrt{\frac{1}{T} \sum_{t=t_0}^T (x_t - \mu)^2} \quad (17)$$

in the formula, σ is the standard deviation of the index, x_t is the instantaneous value of power or speed at time t , and μ is the average value of instantaneous power or speed from the start at time t_0 to the end at time t .

The standard deviations of relative input power and relative rotational speed for the above cases are calculated and shown in Fig. 15. The results indicate that an increase in guide vane closing time correlates with a prolonged duration of speed fluctuation, resulting in a higher standard deviation of the rotational speed. Additionally, a slower closing rate of the guide vanes results in greater output fluctuations and higher standard deviations. The appropriate closing time of the guide vane ensures the stabilization of both the input power and the rotational speed in the disturbed unit. For this study, the recommended closing time is 15 s.

3.3.2. Governor parameters of disturbed unit

To investigate the influence of governor parameters on the dynamic characteristics and regulation performance of the disturbed unit, this subsection analyzes the time-domain variations of the unit's input power with the proportional gain (K_p) and integral gain (K_i). These parameters are selected for study because they are the most fundamental control parameters in VSUs, directly determining system response during hydraulic disturbances and representing the parameters most commonly adjusted in actual pumped storage operations. While increasing K_p generally improves both power and speed stability, excessive parameter values can lead to system instability and fluctuations, as evidenced by increased maximum input power in the tests. Therefore, the study ranges of K_p and K_i were limited to reasonable intervals to ensure a proper balance between enhanced regulation capability and system stability. The results are presented in Fig. 16. The simulation conditions include the guide vane for the disturbing unit closing at a rate of 4% per second and the disturbed unit operating under normal pumping modes with various governor parameters.

Fig. 16(a) and (b) represent the effects of K_p and K_i , respectively. As shown in the figure, K_p and K_i are positively correlated with the maximum value of the input power. For every 0.2 increase in K_p , the maximum input power rises by 0.05%; Similarly, each 0.2 increase in K_i results in a 0.04% rise. K_p and K_i are negatively correlated with the amplitude of input power fluctuations, with K_i having a more significant impact on the amplitude. Additionally, the power terminal dynamics do not show significant variation across various K_p values and can be rapidly stabilized. In contrast, fluctuations in the terminal dynamics are

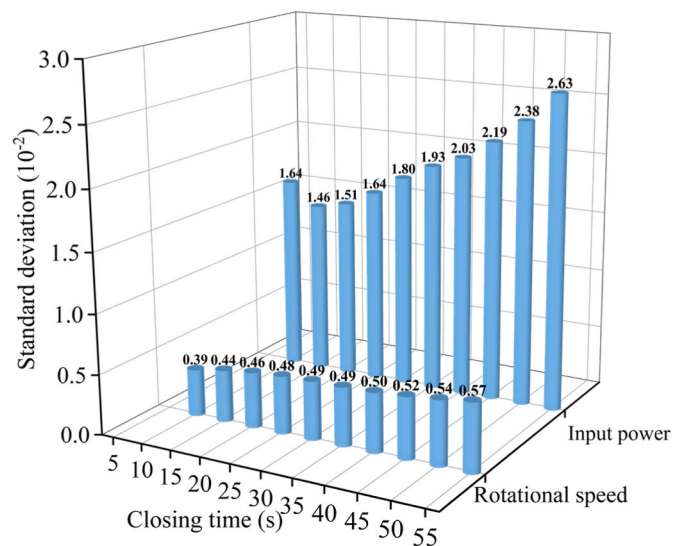


Fig. 15. Standard deviation of speed and power for the disturbed units with different guide vane closing times.

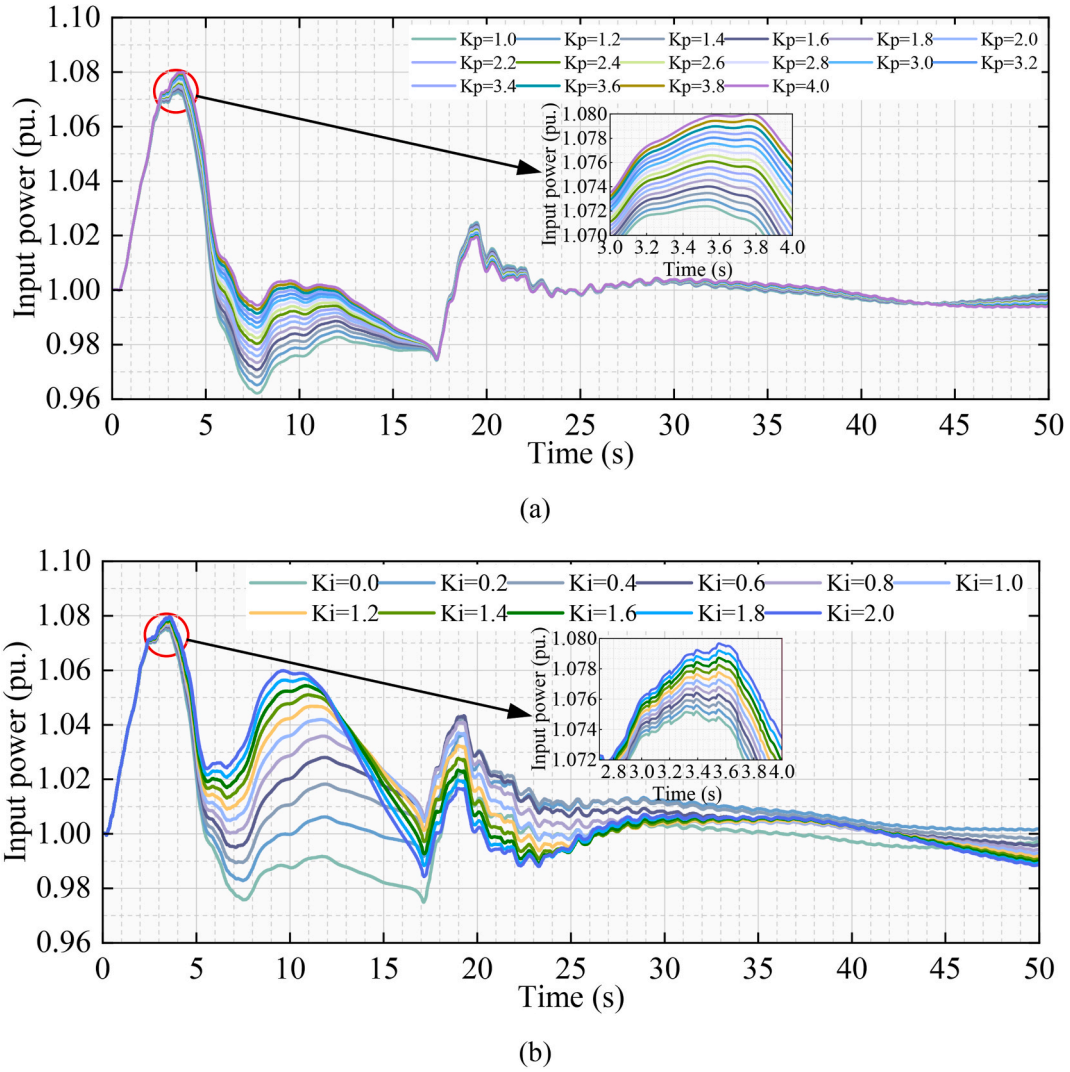


Fig. 16. Time-domain response of input power of the disturbed unit under different control parameters.

considerably different at varying K_i values. These phenomena suggest that the governor parameters significantly influence the power characteristics of the disturbed unit. To maintain stable input power to the disturbed unit, a larger K_p and a smaller K_i are advisable.

VSU leverages its characteristic of variable speed and constant

frequency to alter the input power by adjusting the rotational speed under pumping modes. To explore the correlation between variations in rotational speed and input power, Fig. 17 statistically presents the standard deviations for both the power and the rotational speed of the disturbed unit (Inspired by the data analysis methods from Refs. [36]

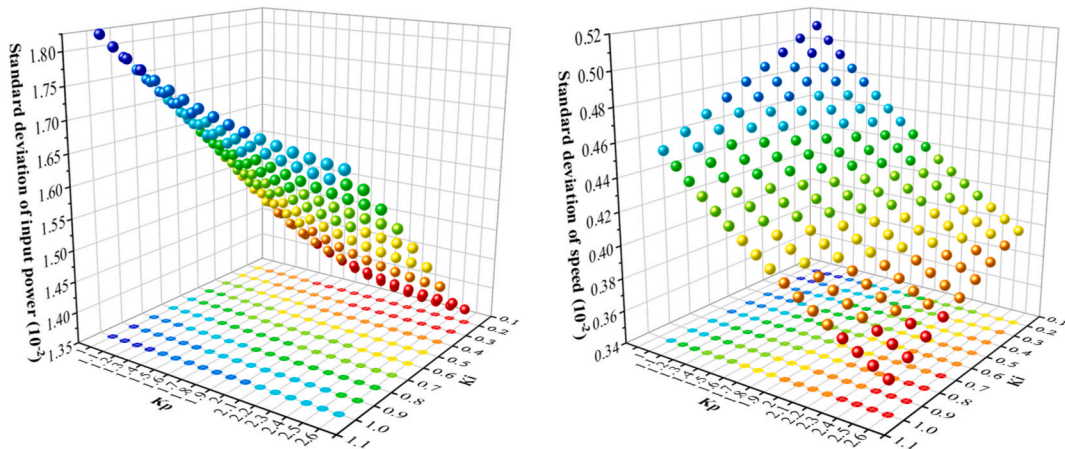


Fig. 17. Standard deviations of power and rotational speed of the disturbed unit under different control parameters.

and [37]). The left figure shows the standard deviation of input power and the right figure shows the standard deviation of rotational speed. It can be observed that larger standard deviations for power and rotational speed are concentrated in the upper right and lower right corners, respectively. This suggests that a higher K_i and lower K_p can contribute to the stabilization of the input power. Conversely, higher values of both K_p and K_i are associated with greater rotational speed stability. To achieve better power stability, it is recommended to use a higher value of K_p while selecting a lower value of K_i , which means a compromise in rotational speed stability. However, this trade-off is reasonable, considering that VSU typically has a high tolerance for minor rotational speed fluctuations.

4. Impact of design parameters - unit combination on VSU hydraulic disturbances

In Section 3, an analysis is conducted on the impact of specific governor parameters, such as the K_p and K_i , on the pump power-off hydraulic disturbances of the VSU. The VSU achieves the highest efficiency under pumping modes through optimal control strategies of the rotational speed. Its speed control system differs from that of the FSU, leading to distinct pump power-off transient processes. This section focuses on a single-pipe dual-machine system to explore the dynamic responses and differences in regulation performance when the VSU and FSU are disturbed.

4.1. Schemes of unit combinations

The configuration of the units is depicted in Fig. 18. Combination (a) consists of two FSUs, while combination (b) consists of one VSU and one FSU. In both combinations, FSU #2 is used as the disturbing unit.

To deeply analyze the dynamic characteristics of the disturbed unit during the transition in different working conditions, this subsection selects two common forms of disturbance for study: pump power-off and pump start-up. The pump start-up process consists of two phases: rotational speed ramp-up and guide vane opening. The rotational speed ramp-up phase primarily involves electrical processes and does not involve hydraulic effects, so it is not discussed in this paper. The explanations for the two disturbance conditions are as follows:

- (1) Pump power-off disturbance: The two units operate at rated input power under the conditions of normal storage levels in the upstream and downstream reservoirs. Unit 1 operates with normal pumping and remains connected to the grid during the disturbance process, with the guide vanes held stationary to minimize reverse hydraulic disturbances in Unit 2. Unit 2 experiences the pump power-off disturbance, adopting a guide vane closing rate of 4% per second (The guide vane closing rate of 4% per second is derived from experimental reports of actual engineering cases used in this study.).
- (2) Pump start-up disturbance: The initial conditions are the same as those of the power-off disturbance, with Unit 1 operating

normally and remaining connected to the grid. Starting from the rated rotational speed with fully closed guide vanes, Unit 2 ramps up to the rated input power at a guide vane opening rate of 4% per second. Table 1 lists the initial and final state parameters of the disturbing unit under various disturbance conditions.

4.2. Results and improvement of unit combination on VSU hydraulic disturbances

To clearly compare the transient characteristics of the FSU and VSU when serving as disturbed units, Fig. 19 shows the dynamic characteristics of input power and rotational speed of the disturbed units under different unit configurations and disturbance forms. Under both pump power-off and start-up conditions, the VSU demonstrates superior input power stability compared to the FSU. Specifically, under the power-off condition, the VSU has a maximum input power of 1.065p.u. and terminal dynamics of 0.088p.u., while the FSU exhibits 1.136p.u. and 0.153p.u., respectively. Upon the start-up condition, the maximum input power and terminal dynamics of the VSU are 1.001p.u. and 0.042p.u., and those of the FSU are 1.007p.u. and 0.063p.u. The enhanced input power stability of the VSU is attributed to its ability to smooth input power fluctuations by adjusting the rotational speed under pumping modes. To verify the general applicability of this inference, we implemented various guide vane control strategies and studied the resulting indicator responses. The findings are detailed in Table 2.

Table 2 shows that the values of the DIP indicators (DIP5 and DIP10) for the disturbed units significantly decrease with an increasing guide vane closing rate during the pump power-off disturbances. DIP5 and DIP10 represent the durations of the transient process in which the input power exceeds its rated value by 5% and 10%, respectively. This suggests that a faster closing rate improves the input power stability for the disturbed units. The reasons are as follows: An excessive DIP duration represents prolonged motor overload operation, potentially leading to heat generation. When the motor's internal temperature reaches the upper limit of its specified range, it may compromise the safety of the unit's materials, thus affecting the safe operation of power plants. Further statistics reveal that the VSU's DIP5 is on average 3.2 s shorter than that of the FSU, its maximum input power is 0.072 p.u. less, and the terminal dynamics exhibit an average reduction of 6.7%. These findings suggest that VSU can improve the resilience to disturbances through adjustments to rotational speeds. Moreover, the disturbance from pump

Table 1
Disturbing unit parameters of working condition transition.

Types of Disturbances	Operating Point State	Rotational Speed (r/min)	Pumping Flow (m^3/s)	Guide Vane Opening (%)
Pump Power-off	Initial State	-428.6	60.4	40.96
	Terminal State	0	0	0
Pump Start-up	Initial State	-428.6	0	0
	Terminal State	-428.6	60.4	40.96

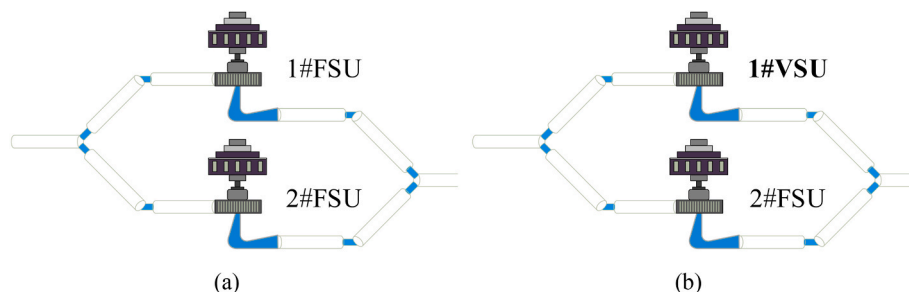


Fig. 18. Combination of units.

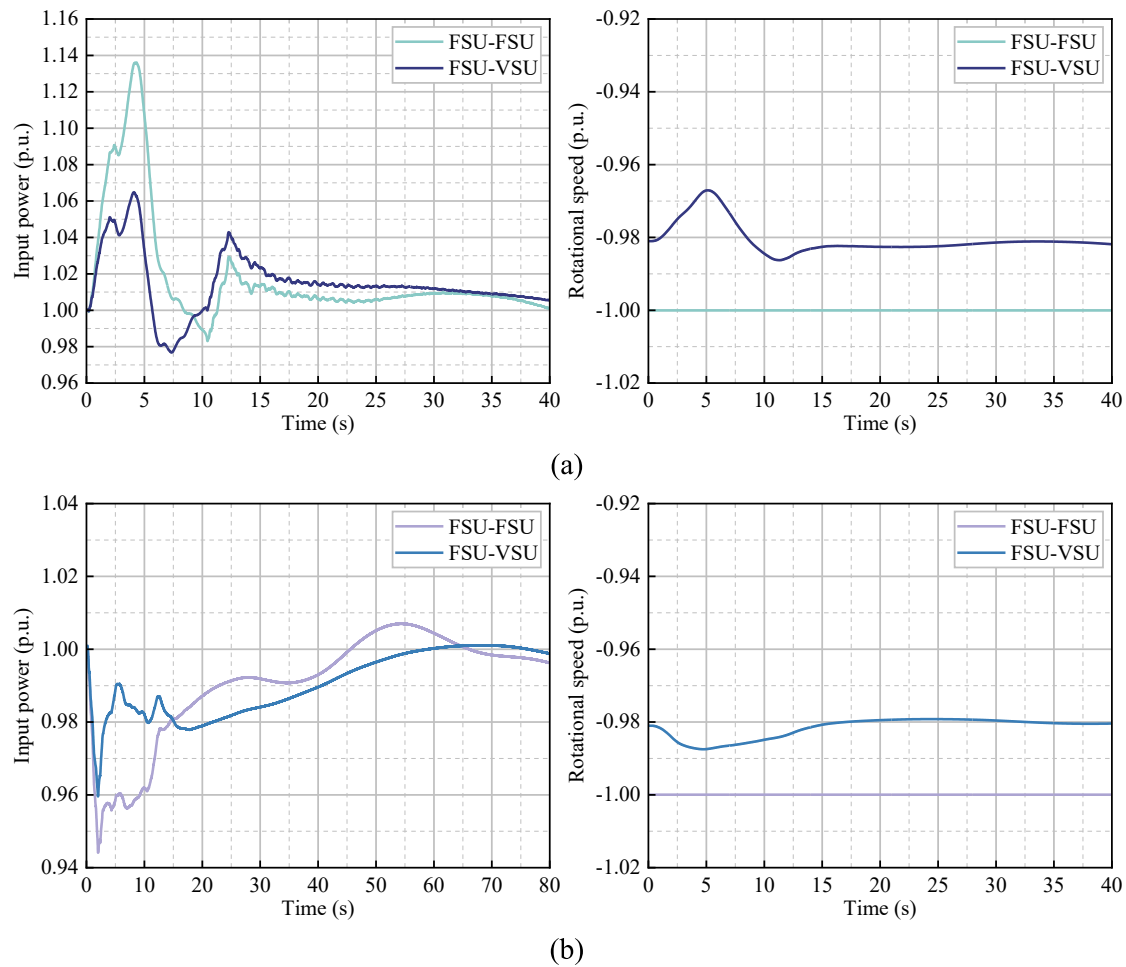


Fig. 19. Dynamic response of the disturbed unit for pump power-off(a) and pump start-up(b).

Table 2

Input power and rotational speed statistics of disturbed units.

Types of Disturbances	Unit Configuration	Guide Vane Action Rate (s)	Maximum Input Power (p.u.)	Input Power Terminal Dynamics (%)	Duration Over 5% of Input Power(DIP5)(s)	Duration Over 10% of Input Power (DIP10)(s)	Rotational Speed Variation (%)
Pump Power-off	Fixed- Fixed	-2%	1.143	16.38	5.2	2.1	0
		-3%	1.141	15.81	4.73	1.97	0
		-4%	1.136	15.30	4.58	1.84	0
		-5%	1.133	15.81	4.47	1.66	0
		-6%	1.126	16.42	4.39	1.34	0
	Fixed -Variable	-2%	1.070	8.07	1.66	0	2.02
		-3%	1.068	8.37	1.48	0	1.98
		-4%	1.065	8.80	1.47	0	1.92
		-5%	1.061	9.57	1.43	0	1.78
		-6%	1.056	11.24	1.26	0	1.52
Pump Start-up	Fixed - Fixed	2%	1.005	3.45	0	0	0
		3%	1.007	4.98	0	0	0
		4%	1.007	6.29	0	0	0
		5%	1.007	7.54	0	0	0
		6%	1.007	8.73	0	0	0
	Fixed - Variable	2%	1.001	2.37	0	0	0.50
		3%	1.001	3.13	0	0	0.68
		4%	1.001	4.15	0	0	0.83
		5%	1.001	5.22	0	0	0.97
		6%	1.001	6.30	0	0	1.10

start-up is significantly less severe than during pump power-off conditions, with the DIP10 value even dropping to zero, indicating that pump start-up does not pose a hazardous condition relative to pump power-off.

In summary, the VSU demonstrates superior resilience to hydraulic

disturbances compared to the FSU under different unit control strategies. Additionally, a faster closing rate of the guide vanes in the disturbing units enhances the input power stability of the disturbed unit. Within the safe guide vane closing range of the disturbing unit, the

operators aim to find the optimal closing rate that ensures the safety of the disturbing unit while maintaining the stability of the disturbed unit. In this case, the optimal closing rate determined by the study is 5%/s and the opening rate is 4%/s.

Fig. 20 show the standard deviation of the input power and rotational speed for the disturbed unit under pump power-off and start-up conditions, respectively. Under the pump power-off condition, the rotational speed deviation of the VSU is negatively correlated with the guide vane closing rate, while the input power deviation shows no significant correlation with this rate. The FSU maintains a constant rotational speed without deviation due to its connection to the grid, but the input power deviation is negatively correlated with the closing rate. Effective control of the rotational speed fluctuation in VSU and the power fluctuation in FSU can be attained by adjusting the closing rate. Additionally, the VSU has an average input power standard deviation of 1.37%, which is less than that of the FSU at 2.43%, demonstrating the significant advantage of the VSU in reducing input power fluctuation during pump power-off disturbances.

Under the pump start-up condition, both the rotational speed and input power standard deviation of the VSU are positively correlated with the guide vane opening rate. Similarly, the input power deviation of the FSU correlates positively with this rate. Studies have shown that a smaller guide vane opening rate enhances the stability of the disturbed unit during pump start-up. The average input power standard deviation of the VSU is 0.88%, which is lower than that of the FSU at 1.43%. This suggests that the VSU's characteristics of variable speed and constant frequency contribute to a reduction in input power fluctuations during pump start-up disturbances.

Overall, appropriate fluctuations in the rotational speed of VSU can effectively mitigate input power fluctuations during pump power-off and start-up conditions. Furthermore, a faster guide vane closing rate helps improve the speed stability of the disturbed unit during pump power-off, while a slower guide vane opening rate helps improve the stability of the disturbed unit during pump start-up. The disturbance intensity during pump start-up is less severe than during power-off conditions.

5. Impact of design parameters - pressure pipeline system arrangements on VSU hydraulic disturbances

In the design and operation of pumped-storage power plants, researchers must consider both the type of unit (variable or fixed speed)

and the structure of the pressure pipeline system to achieve optimal power regulation performance and enhance resilience to hydraulic disturbances. This section explores the effect of pressure pipeline system's structure on pump power-off hydraulic disturbances.

5.1. Types of pipeline system arrangements

Hydraulic disturbances typically occur in multi-unit systems. Due to the shared diversion and draft tunnels, the water hammer pressure may cause flow redistribution at the bifurcated pipes when the working conditions of the disturbing unit are changed and its input power is varied. In this subsection, the impacts of various pipe arrangements on hydraulic disturbances during pump power-off are studied for a common layout where one tunnel serves two units, sharing both diversion and draft tunnels. The study considers the isolation of unilateral disturbances. The specific scenes of the pipe arrangements are shown in Fig. 21.

To ensure that factors such as pipe diameter, length, and arrangement of surge tanks do not affect the study of hydraulic disturbance characteristics under different pipe arrangements, the following assumptions have been made for the draft and diversion sections in Scenes 2 and 3.

- (1) The number and length of the pipes through which each segment of the flow passes in the pressure pipeline system are maintained consistent with those of the original structure.
- (2) To maintain a constant fluid flow velocity, the diameter of the draft pipe in Scene 2 and the diversion pipe in Scene 3 will be adjusted according to the principle of halving the flow rate while keeping the flow velocity unchanged. In Scene 2's draft tunnel and Scene 3's diversion tunnel, the flow rate is reduced to half of the original. To maintain the same flow velocity as in the original structure, the tunnel diameter is adjusted to $1/\sqrt{2}$ of the original.
- (3) The location of the surge tank remains the same as in the original structure. Due to the flow being halved, both the chamber area of the surge tank and the area of the impedance hole are reduced to half of their original size. This means the diameters of the chamber and the impedance hole are reduced to $1/\sqrt{2}$ of their original sizes, in order to maintain the fluid flow pattern as much as possible.

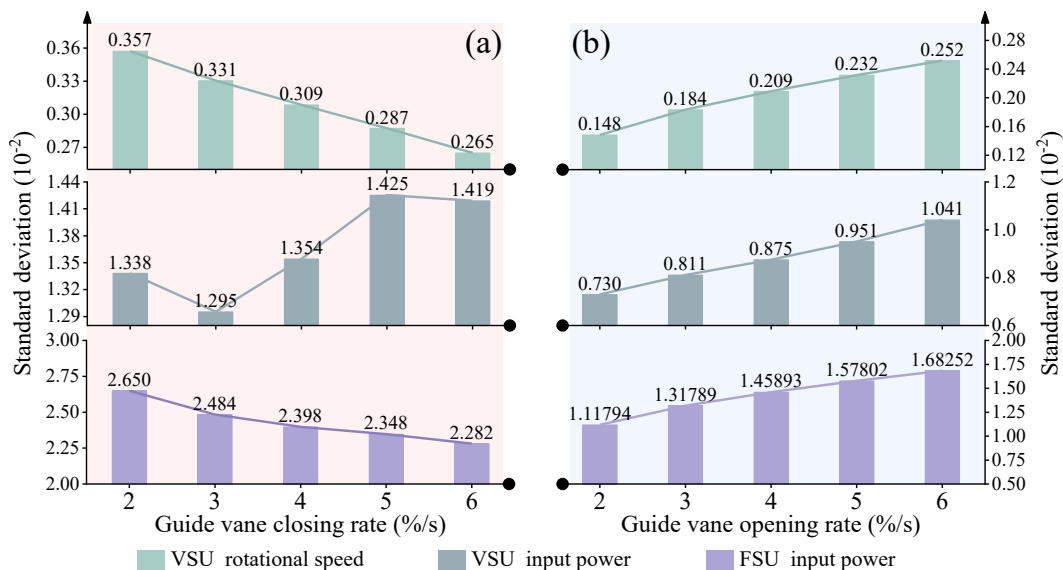


Fig. 20. Standard deviation of input power and rotational speed for the disturbed unit under pump power-off(a) and start-up(b) conditions.

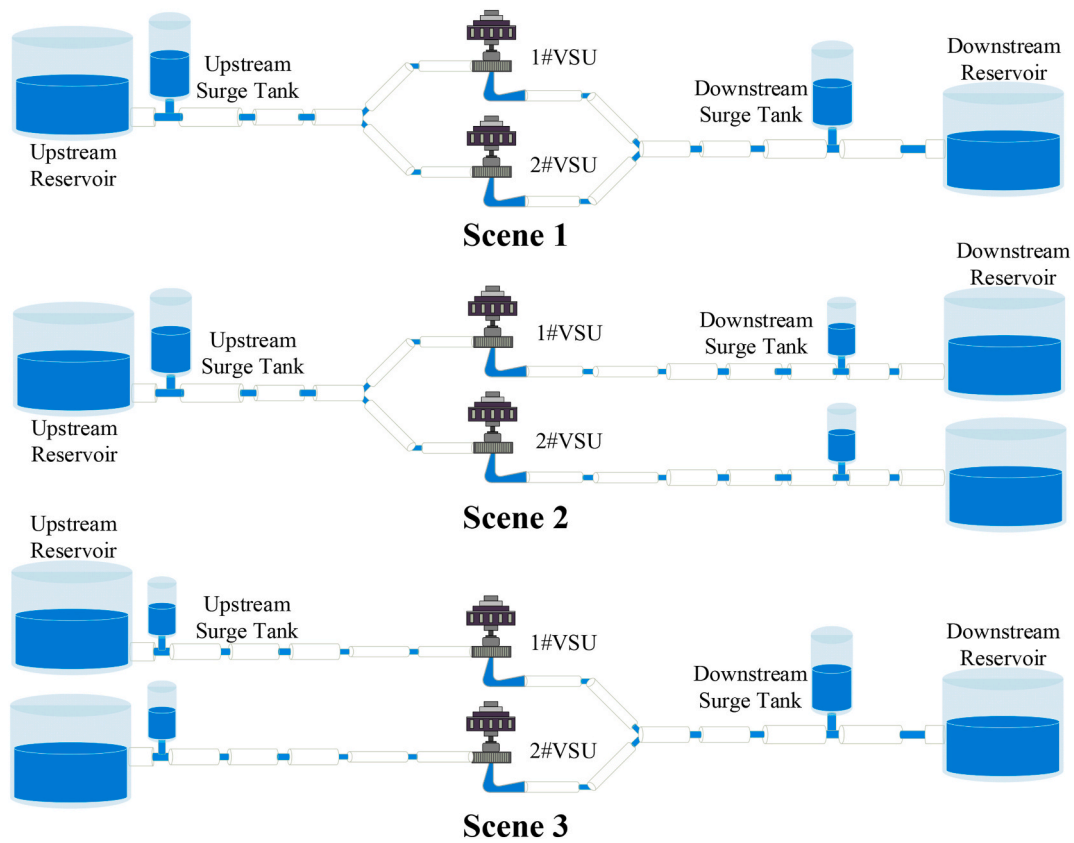


Fig. 21. Arrangements for unilateral and bilateral disturbances.

5.2. Results and improvement of pipeline system arrangements on VSU hydraulic disturbances

To investigate the impact of pipe arrangement on the pressure characteristics of hydraulic disturbances, the optimal steady-state parameters from Section 3.1 are utilized. With the guide vane closing time set to 30 s, pump power-off disturbance transient process is simulated across three different pipe arrangement scenes. Unit 1 is considered the disturbed unit, while Unit 2 acts as the disturbing unit. For Unit 1, pressure changes result from two primary influences: First, Unit 1 is disturbed by the water flow transmitted through the shared pipeline from Unit 2; second, there are pressure fluctuations within Unit 1 itself.

Fig. 22 shows the dynamic response of the pressure in the disturbing and disturbed units under various pipe arrangement forms. Under Scene 1, which involves a shared diversion and draft pipe, the pressure trend for Unit 1 is consistent with that for Unit 2, suggesting synchronized pressure changes from the shared pipe. Due to the energy loss from the water hammer wave, Unit 1 will not encounter stability issues when the pressure extremes in Unit 2 remain within safe limits. Under Scene 2, with only a shared diversion, the pressure changes in the spiral case of Unit 1 are consistent with those of Unit 2. However, the draft pipe pressure changes in Unit 1 are smaller, indicating that the draft pipe pressure in Unit 1 is primarily influenced by the internal spiral case pressure changes, rather than the draft pipe disturbances from Unit 2. In Scene 3, where only a shared draft pipe is present, the draft pipe pressure of Unit 1 is influenced by Unit 2, but the changes in the spiral case pressure are insignificant. The minimal changes can be attributed to the effect of the shared draft pipe, where the water hammer pressure tends to propagate primarily upstream, thereby reducing the impact on the spiral case of Unit 1. To more intuitively observe the impact of the three structures on the unit's pressure-speed-flow characteristics, the following section illustrates the characteristic trajectories of the disturbed unit during the regulation process.

Fig. 23 (a), (b), and (c) show the pressure time-domain diagrams and rotational speed-flow characteristic trajectories of the disturbed unit under Scene 1, Scene 2, and Scene 3, respectively. The time-domain diagrams illustrate the unit's pressure variations over time, where the black circle represents the start of regulation point a, the square represents the minimum pressure point b, and the triangle represents the maximum pressure point c. In (a), the second plot shows the rotational speed-flow trajectory from a to b, and the third plot shows the rotational speed-flow trajectory from b to c; (b) and (c) are similar. In the trajectory plots, the horizontal axis represents relative rotational speed, and the vertical axis represents relative flow, with warmer colors indicating high-efficiency regions and cooler colors indicating low-efficiency regions. The figures indicate that during the adjustment process, the unit operates along the edge of the characteristic curve in the highest efficiency region, with minimal guide vane adjustments, facilitating rapid regulation. Additionally, the range of changes in speed and flow in Scene 2 is slightly smaller than in Scene 1, while Scene 3 shows a range much smaller than Scene 2. This suggests that the disturbances to the unit are primarily caused by the shared diversion pipe. Since the water flow carries shock forces upstream, disturbances caused by the shared draft pipe are significantly less than those caused by the shared diversion pipe. The arrangement in Scene 3 isolates the impact of the shared diversion water on the spiral case and eliminates disturbances to the unit's flow and rotational speed. This configuration reduces the effect on the unit's pressure, rotational speed, and flow by only transmitting pressure through the draft water and unit internal mechanisms.

Considering the effect of guide vane closing time on hydraulic disturbance, the study investigates the extremal in water head of the disturbed units under different guide vane closing times and pipe arrangements, as shown in Fig. 24. The results show that as the guide vane closing time increases, the maximum water heads in Scenes 1 and 2 exhibit a trend of initially decreasing and then increasing with the closing time, with average values of 615.3 m and 615.7 m, respectively.

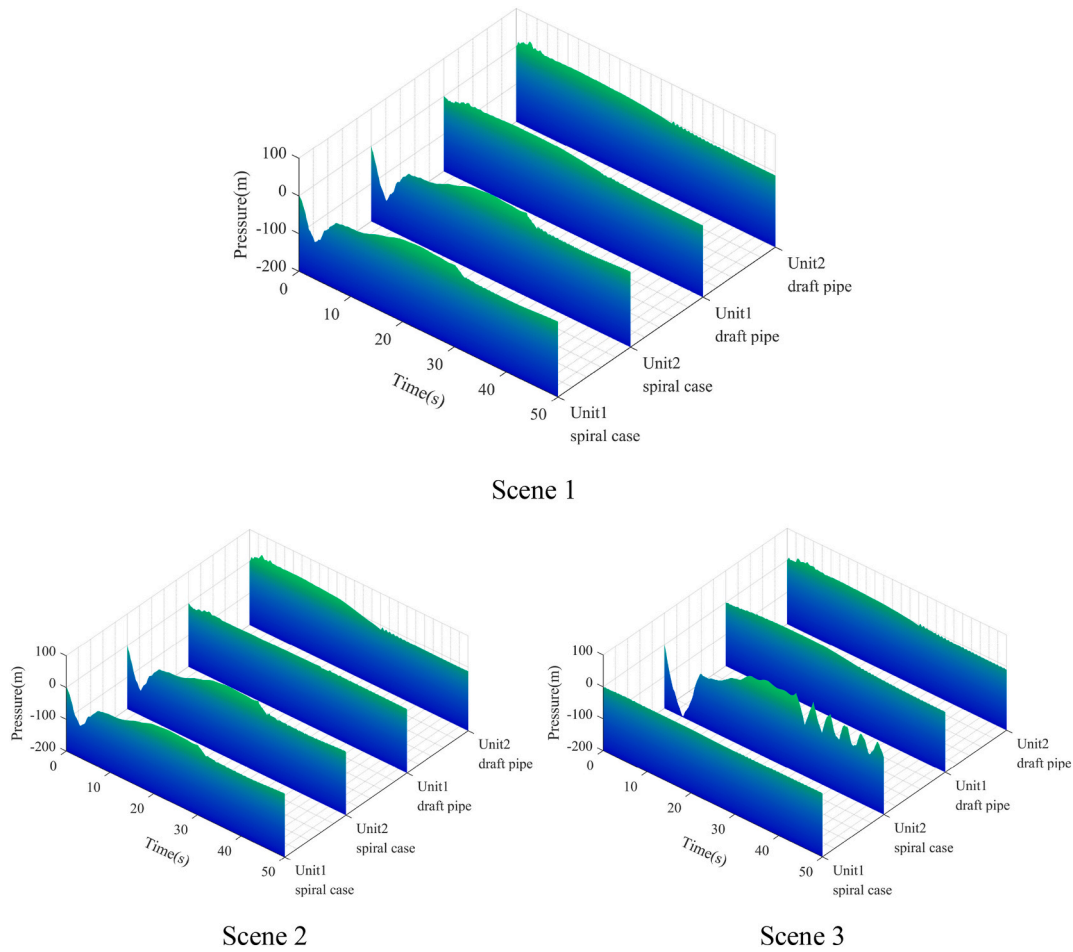


Fig. 22. Dynamic response of the pressure in the disturbing and disturbed units under various pipe arrangement forms.

The maximum water head in Scene 3 is positively correlated with the guide vane closing time with an average value of 559.1 m, which is significantly lower than the previous two scenes. Bar chart of Fig. 24 represents the water head variations in different arrangements, reflecting the degree of water head fluctuation. The water head variations in Scenes 1 and 2 first increase and then decrease with the closing time, while those in Scene 3 increase linearly with the closing time. The average water head variations for Scenes 1, 2, and 3 are 130 m, 122.1 m, and 21.3 m, respectively, with a much smaller variation in Scene 3 compared to Scenes 1 and 2.

The above phenomenon occurs because the disturbing unit operates in pumping mode, and the water hammer energy generated during the power-off process is mainly transferred upstream by the diversion pipe. In Scene 3, the two units do not exchange fluids through the diversion pipe, resulting in small water head fluctuations that are relatively stable over time. In contrast, the shared diversion pipe between Scenes 1 and 2 results in greater water head variations.

Table 3 demonstrates the variations in the rotational speed and input power standard deviation of the disturbed unit across different pipe arrangements. The order of standard deviation for rotational speed and input power is Scene 1 > Scene 2 > Scene 3, indicating that Scene 1, featuring a shared diversion and draft pipe, has greater disturbances on the two parameters. This impact is attributed to the water hammer wave from the disturbing unit affecting the disturbed unit through both the diversion and draft pipes. Scenes 2 and 3 mitigate the water hammer effect by reducing unilateral disturbances. Furthermore, the standard deviations rise with the increase in the closing time of the guide vane in the disturbing unit. A shorter closing time of the guide vane results in less flow along the pipe, generating a weaker water hammer wave and a

reduced impact on the disturbed unit. Therefore, the fluctuations in the input power and rotational speed of the disturbed unit diminish as the closing time of the guide vane decreases.

Over all, in the construction of variable speed PSPP, selecting an independent pipe arrangement benefits stable unit operation if permitted by the project investment. Furthermore, under different structural configurations, a higher guide vane closing rate during a pump power-off disturbance can enhance power stability, but it may worsen pressure stability. In practical operations, a balance should be struck.

6. Conclusion and discussion

6.1. Conclusion

The adjustable power of the VSU in pumping modes is one of its key advantages, and its transient characteristics are essential to ensure the stable absorption of active power on the grid. To enhance the hydraulic disturbance characteristics, this paper establishes a pumping system model for PSPP, evaluates key parameters, and proposes improvement measures. The following are the main conclusions drawn from the study:

- (1) Comprehensive evaluation of hydraulic disturbance due to VSU pump power-off
 - Operating parameters

When the VSU encounters the same disturbance at different initial operating rotational speeds, a higher initial speed results in increased spiral case pressure and instantaneous flow in the disturbing unit, while

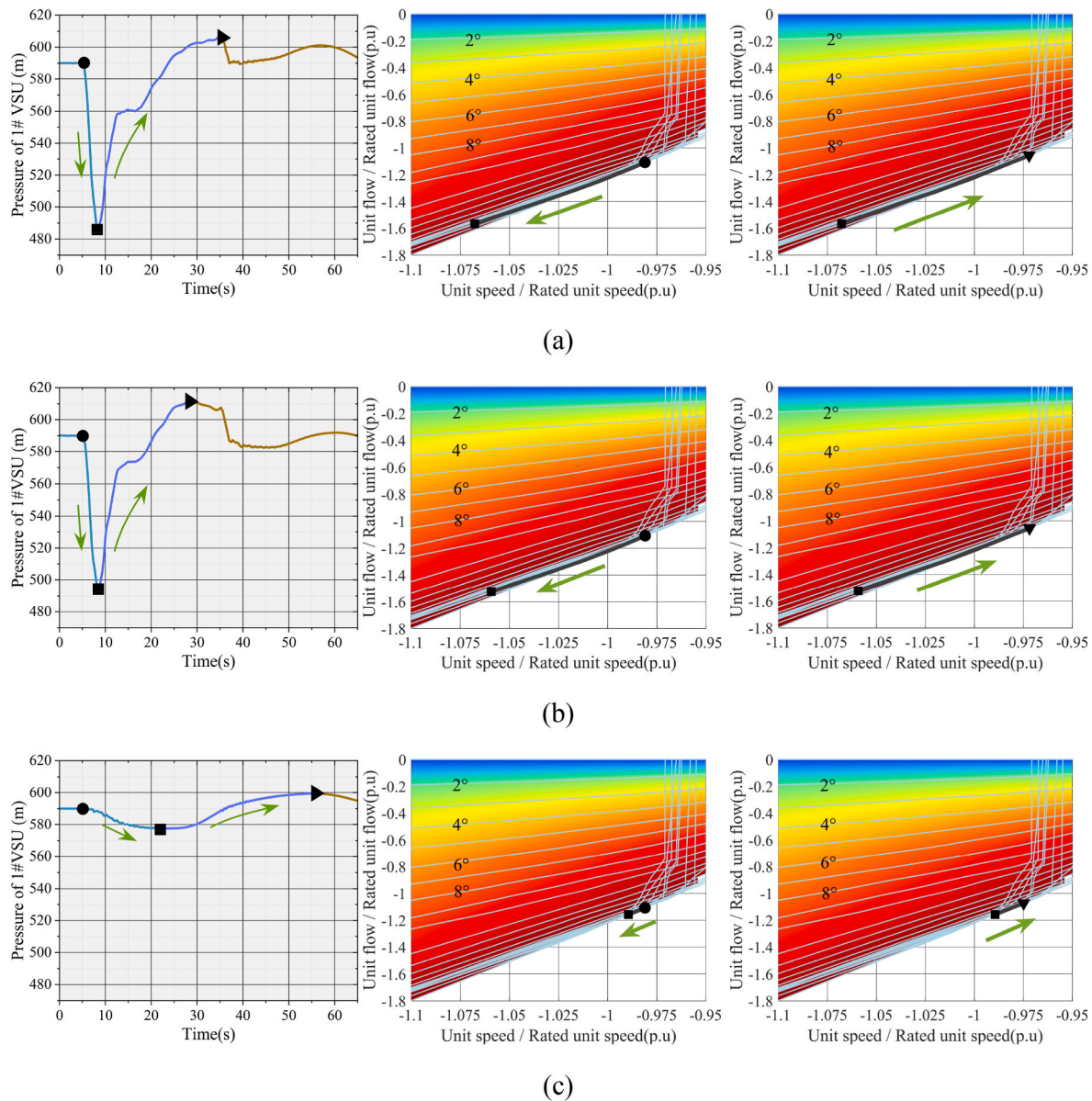


Fig. 23. Rotational speed-flow characteristic trajectories of the disturbed unit under different pipeline structures.

the flow characteristics of the disturbed unit improve.

● Design parameters

Comparing the pump power-off disturbances between the VSU + FSU and FSU + FSU schemes, it is observed that under the same disturbance conditions, the VSU maintains shorter durations of input power exceeding 5% and 10%, demonstrating a stronger recovery capability. However, the VSU exhibits more pronounced speed fluctuations than the FSU. Within the safety range, a larger guide vane closing rate helps to mitigate these fluctuations.

Assess the impact of three layout schemes—shared diversion pipe, shared draft pipe, and shared diversion and draft pipes—on transient performance. The results indicate that when unilateral disturbances are isolated, the changes in spiral case pressure of the disturbed unit are primarily caused by the shared diversion pipe, with minimal impact from the draft pipe. Additionally, the layout with only shared draft pipe provides smaller variations in flow and speed for the disturbed unit.

(2) Measures to enhance resistance to hydraulic disturbance

● Operating aspects

The speed governor parameters have a significant impact on the input power and speed stability of the disturbed unit. Increasing K_p helps to maintain the stability of input power and speed, while increasing K_i is not beneficial for input power stability but is advantageous for speed stability.

● Design aspects

The actuation rate of the guide vane within the safe range has the potential to enhance the stability of the disturbed unit. Specifically, increasing the closing rate of the guide vane for the disturbed unit helps to improve the stability of the input power for the disturbed unit during pump power-off, while reducing the opening rate of the guide vane for the disturbed unit aids in enhancing the stability of the disturbed unit during pump start-up.

When investment permits, a shared draft pipe layout can effectively enhance the unit's resistance to hydraulic disturbance. Among the three layout schemes, the shared draft pipe layout has the smallest average

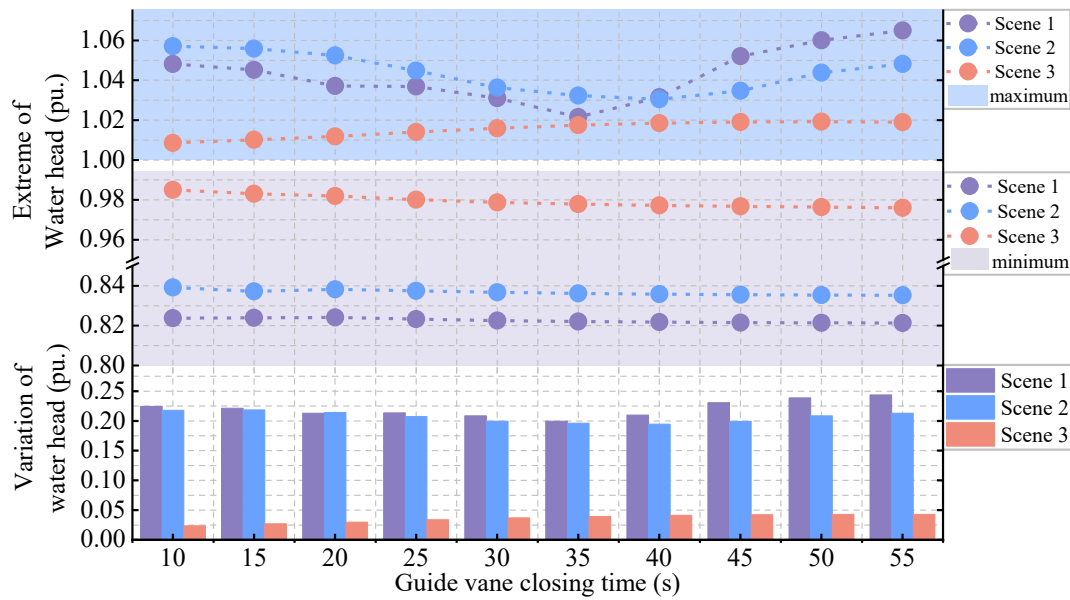


Fig. 24. Comparison of water head extremes of units with different arrangement forms.

Table 3

Standard deviation of rotational speed and input power for different pipe arrangements.

Index Standard deviation Closing time	Rotational speed			Input power		
	Scene 1	Scene 2	Scene 3	Scene 1	Scene 2	Scene 3
10	0.39	0.40	0.10	1.72	1.62	0.51
15	0.43	0.44	0.12	1.55	1.34	0.56
20	0.46	0.47	0.14	1.60	1.34	0.64
25	0.48	0.48	0.16	1.74	1.38	0.72
30	0.49	0.47	0.18	1.89	1.44	0.81
35	0.49	0.46	0.19	2.02	1.49	0.88
40	0.50	0.45	0.20	2.13	1.54	0.93
45	0.51	0.46	0.20	2.31	1.64	0.96
50	0.54	0.48	0.21	2.54	1.77	0.97
55	0.58	0.50	0.21	2.83	1.95	0.98
Average	0.487	0.46	0.1702	2.0312	1.549	0.7948

standard deviation of input power. In the case study presented in this paper, the shared draft pipe layout improves speed stability by at least 63% and power stability by at least 48.7% compared to the other two schemes.

6.2. Discussion

6.2.1. Limitations of the proposed evaluation framework

VSU technology started relatively late in China, and there is a lack of related standards and design guidance. As the adjustability of the pump operating conditions is a major advantage of VSU [29], we have reason to believe that future standards will include information regarding hydraulic disturbances caused by pump power-off. Additionally, the three-dimensional evaluation framework presented in this paper aligns with broader trends in energy system optimization research. Similar multi-parameter optimization approaches have proven effective in other energy applications, such as enhancing building energy performance

through systematic evaluation of design parameters [38], and this methodological convergence across energy fields demonstrates the universal applicability of comprehensive evaluation frameworks in achieving sustainable energy goals. Nevertheless, there are still deficiencies.

Given that two-unit systems are the most common configuration in practical engineering, this study focused on the evaluation and optimization of two-unit systems. However, the proposed three-dimensional evaluation framework inherently possesses good scalability and can be applied to three-unit or four-unit systems. The core principles of the framework, namely analyzing control strategies, unit combinations, and structural arrangements, are equally applicable to larger-scale systems. It should be noted that multi-unit systems introduce additional complexities, including complex hydraulic interference coupling effects, the formulation of coordinated speed control strategies, and the diversified design of penstock and draft tube system arrangements. These all require more refined coordination strategies, and the corresponding

computational time will increase significantly. Therefore, developing this evaluation framework into a visual tool for easier operation and application will offer broader application prospects in the future. It should be noted that this study focuses on short-term transient process analysis (seconds to minutes) at the equipment level, primarily providing technical support for unit design optimization and operational control. While long-term sustainability assessment involves different time scales and research scope, the reliable technical evaluation framework established here provides an important foundation for subsequent long-term impact studies.

Additionally, the framework presents several practical implementation constraints that warrant consideration. The three-dimensional evaluation approach requires extensive computational resources and detailed hydraulic and control system parameters, which may not be readily available at all pumped-storage facilities. The framework's reliability is contingent on accurate initial condition data and site-specific calibration of control strategy parameters. Furthermore, while the core principles are applicable across different VSU configurations, the current validation is primarily based on doubly-fed variable-speed units, and results may require adaptation for other VSU technologies.

In the study of the three structures in this paper, assumptions were made about the position of the surge tank and pipe diameter in order to avoid the influence of other structures on the results. In the future, the impact and patterns of structural parameters of variable speed units on pump power-off, load rejection, frequency regulation, and other operating conditions will be explored, providing more detailed and valuable insights.

6.2.2. Framework adaptability to different operational contexts and scenarios

The framework demonstrates excellent scalability for different system configurations. While this study focuses on two-unit systems as the most common configuration in practical engineering, the three-dimensional evaluation approach can be extended to three-unit, four-unit, or larger multi-unit systems. The evaluation methodology remains consistent across different scales, with the primary adjustments involving parameter calibration and computational resource allocation. For larger systems, the framework can accommodate increased hydraulic interference coupling effects and more complex coordinated control strategies by expanding the control strategy dimension, including multi-unit coordination algorithms and load distribution optimization.

The framework's applicability extends beyond pump power-off disturbances to encompass various operational scenarios including load rejection, frequency regulation, and emergency shutdowns. For load rejection scenarios, the evaluation framework can modify performance indicators to focus on hydraulic transients. During frequency regulation operations, the framework can emphasize speed governor response characteristics and power tracking capabilities. The three-dimensional structure allows for scenario-specific weighting of evaluation criteria while maintaining the comprehensive assessment approach.

Appendix A

Boundary conditions for the pipeline.

(1) Upstream and downstream reservoir

6.2.3. Implementation challenges and practical considerations

The computational demands of the three-dimensional evaluation approach represent a significant challenge, as the framework requires substantial processing power and data storage capabilities to handle continuous monitoring and analysis. The computational time increases substantially with system complexity, particularly for multi-unit configurations, which requires facilities to ensure adequate computational infrastructure while maintaining real-time operational responsiveness.

Operational implementation involves extensive site-specific calibration of control strategy parameters and establishment of accurate baseline data, which depends heavily on the quality and consistency of input information that may be challenging to maintain in practical environments. Operators require comprehensive training to effectively utilize the framework and interpret its outputs for operational decision-making, while the framework's reliability remains contingent on maintaining high-quality data inputs and regular system validation.

Economic considerations include significant deployment costs encompassing hardware upgrades, software implementation, operator training, and ongoing maintenance requirements, necessitating careful cost-benefit analysis given that the framework's primary benefits may be most evident during disturbance events rather than normal operations. Regulatory compliance represents an additional consideration, as implementation must align with existing grid codes and safety regulations, while any recommended control parameter adjustments require validation against regulatory requirements and potential approval from grid operators before operational deployment.

CRediT authorship contribution statement

Yu Xiao: Writing – original draft, Visualization, Methodology, Conceptualization. **Liuwei Lei:** Writing – review & editing, Methodology, Data curation. **Xinmeng Ma:** Writing – review & editing, Data curation. **Yongyan Chen:** Writing – review & editing, Methodology, Data curation. **Yuwen Deng:** Methodology, Investigation. **Ziwen Zhao:** Validation, Data curation. **Diyi Chen:** Supervision, Resources, Project administration. **Ye Zhou:** Writing – review & editing, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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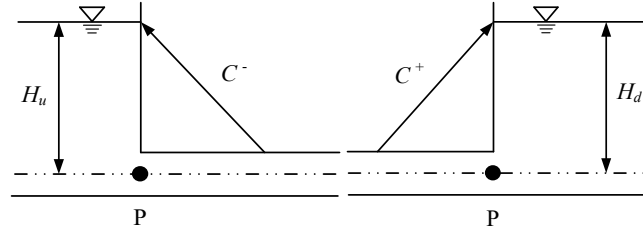


Fig. A.1. Upstream and downstream reservoir

In PSPP, the reservoir level is considered to be a constant in the simulation because it changes slowly. By incorporating an auxiliary equation, the relationship among H_p , H_u , and H_d can be expressed based on the energy equation [21].

$$C^- : H_p = C_M - B_M Q_p \quad (A.1)$$

$$H_p = H_u - \zeta \frac{Q_p |Q_p|}{2gA^2} \quad (A.2)$$

$$C^+ : H_p = C_P - B_P Q_p \quad (A.3)$$

$$H_p = H_d + \zeta \frac{Q_p |Q_p|}{2gA^2} \quad (A.4)$$

where ζ is the local head loss coefficient; H_p is the pressure at node P in Fig. A.1. The simultaneous Eqs. (A.1), (A.2), (A.3), and (A.4) can be solved to determine the pressure and flow at the upstream and downstream reservoirs.

(2) Series pipe

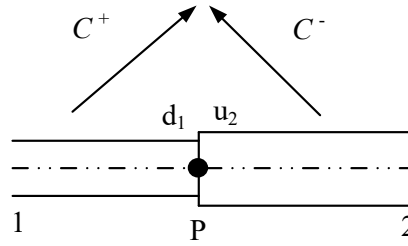


Fig. A.2. The series pipe

Fig. A.2 depicts a series pipe node where Pipes 1 and 2 have different properties, and compatibility equations are established for each side:

$$C^+ : H_p = C_P - B_P Q_p \quad (A.5)$$

$$C^- : H_p = C_M - B_M Q_p \quad (A.6)$$

As a transition between the main and branch pipes, the series pipe presents two additional degrees of freedom, necessitating the introduction of two auxiliary equations. One is the continuity equation, which indicates that the flow is equal on both sides of the node. The other is the energy equation, showing that the head on the left side of the node equals the head on the right side of the node minus the head loss occurring through the node, as detailed in Eqs (A.7) and (A.8):

$$Q_{d1} = Q_{u2} = Q_p \quad (A.7)$$

$$H_{d1} = H_{u2} + \zeta \frac{Q_p |Q_p|}{2gA^2} \quad (A.8)$$

Simultaneous equations (A.5), (A.6), (A.7) and (A.8) can be solved to determine the various parameters at the node of a series pipe.

(3) Bifurcated pipe

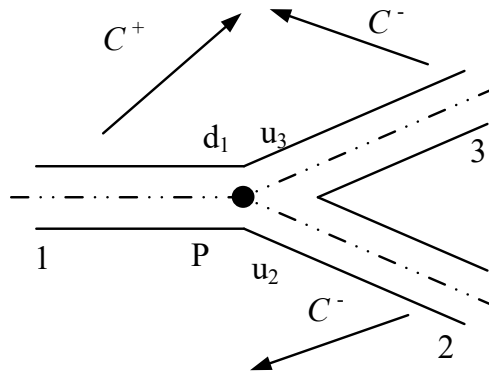


Fig. A.3. The bifurcated pipe

As the bifurcated pipe node shown in Fig. A.3, water flows from Pipe 1 and splits into Pipes 2 and 3 at Point *P*. The C^+ equations can be formulated for section d_1 , and the C^- equations can be formulated for sections u_2 and u_3 :

$$C^+ : H_{p1} = C_{p1} - B_{p1} Q_{p1} \quad (\text{A.9})$$

$$C^- : H_{p2} = C_{M2} - B_{M2} Q_{p2} \quad (\text{A.10})$$

$$C^- : H_{p3} = C_{M3} - B_{M3} Q_{p3} \quad (\text{A.11})$$

The node of the bifurcated pipe exhibits three additional degrees of freedom, necessitating the introduction of three auxiliary equations for a comprehensive analysis. According to the continuity equation, the flow on the left side of the node is equal to the sum of the flows in the two pipes on the right side of the node. Assuming minor head loss at the fork, the energy equation states that the head on the left side of the node is equal to the combined head of the two pipes on the right side. As shown in the following equations:

$$Q_{p1} = Q_{p2} + Q_{p3} \quad (\text{A.12})$$

$$H_{p1} = H_{p2} \quad (\text{A.13})$$

$$H_{p1} = H_{p3} \quad (\text{A.14})$$

Simultaneous equations (A.9) - (A.14) can be solved to determine the various parameters at the bifurcated pipe node.

(4) Surge tank

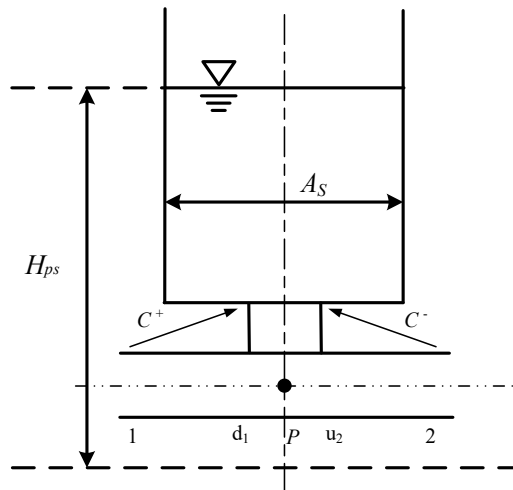


Fig. A.4. The surge tank

As shown in Fig. A.4, for Pipes 1 and 2, the compatibility equations that exist for both sides are as follows [16]:

$$C^+ : H_{p1} = C_{p1} - B_{p1} Q_{p1} \quad (\text{A.15})$$

$$C^- : H_{p2} = C_{M2} - B_{M2} Q_{p2} \quad (\text{A.16})$$

The surge tank serves as the endpoint of the two pipes, and two auxiliary equations need to be added. The first is the continuity equation, which states that the flow to the left of point *P* is the sum of the flows to the right of point *P* and the flow into the surge tank. The second is the energy equation, where the head at point *P* is equal to the heads to the left and right of it.

The flow into the surge tank needs to be determined based on the characteristic equations specific to each type of surge tank. For instance, the equations for the impedance surge tank is as follows:

$$H_p = H_{ps} + R_s Q_{ps} |Q_{ps}| \quad (\text{A.17})$$

$$H_{ps} = H_{ps0} + \frac{(Q_{ps} + Q_{ps0}) \Delta t}{2A_s} \quad (\text{A.18})$$

where H_{ps} and H_{ps0} represent the water levels in the surge tank at the current time and at the end of the last time period, respectively; Q_{ps} and Q_{ps0} are the flows in the surge tank at the current time and at the last time period, respectively; A_s is the cross-sectional area of the open chamber in surge tank; R_s is the hydraulic loss coefficient of the impedance hole in the surge tank. The parameters of the impedance surge tank node can be solved by the simultaneous equations (A.15)–(A.18).

Appendix B

B.1 Hydraulic system validation

In this section, a pumped storage power plant in China, currently operational and equipped with four 300 MW mixed-flow pumped storage units, is selected for analysis to verify the accuracy of the simulation model. A simulation model is constructed based on the pipe and unit data of this power plant. The operating parameters of the power plant are as follows: an upstream normal storage level of 733 m, a downstream normal storage level of 181 m, a rated net water head of 540 m, and a rated flow of 62.56 m³/s. The power-off process of a dual-unit pump is simulated using the proposed model, and the simulation results are compared and analyzed with the measured data. The time-domain response comparison of the characteristic parameters between the simulated and measured data is in Fig. B.1. The specific parameter error comparison data are summarized in Table B.1.

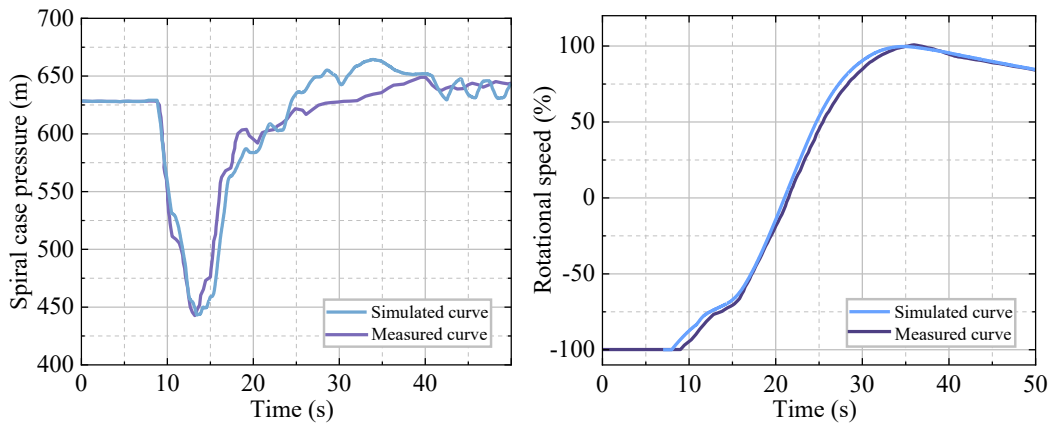


Fig. B.1. The comparison between the measured trend line and the simulation curve

Fig. B.1 demonstrates a comparison of the measured and simulated curves for the spiral case pressure and the relative rotational speed of the unit during a dual-unit pump power-off process. The analysis results show that the simulation curve of the spiral case pressure generally corresponds to the measured trend line, despite minor deviation at the maximum value. In contrast, the simulation curve for the relative rotational speed closely matches the trend line of the measured data. Additionally, we validated the accuracy of the draft tube simulation in our model in paper [16].

Table B.1

The error comparison between measured and simulated values

Parameter	Measured Value	Simulated Value	Error (%)
Pre-Power-off Net Head (m)	567.38	567.38	0
Initial Spiral Case Pressure (m)	628.74	629.05	0.049
Maximum Spiral Case Pressure (m)	649.53	664.38	2.29
Minimum Spiral Case Pressure (m)	442.71	443.2	0.11
Speed Reversal Extremes (rpm)	100.57	99.71	0.681
Time of Speed Reversal Extremes (s)	26.45	26.27	0.854

Table B.1 presents a comparison between the measured and simulated values for parameters such as pre-power-off net head, initial spiral case pressure, and rotational speed reversal extremes during the model validation process. Among the listed parameters, initial spiral case pressure exhibits the largest error at 2.29%. However, the simulated value exceeds the measured one, which is more favorable for the safety and stability analysis of the actual project. The errors of other parameters are less than 1%. With an average error of 0.66% for the listed parameters, the simulation curve is essentially consistent with the measured trend line. This demonstrates the accuracy and reliability of the simulation model developed in this paper and satisfies the standards required for practical engineering safety.

B.2 Control and electrical system validation

To verify the accuracy of the control system and electrical subsystem models established in this paper, this section focuses on examining the transient response characteristics of the excitation system under power regulation mode, including the rapidity and precision of regulation, with the unit speed maintained at -0.98 p.u. To validate the reliability of the model, the simulation results are compared and analyzed with real machine experimental data from Ref. [39]. The experimental design includes two sets of power reduction command conditions (T1, T2) and two sets of power increase command conditions (T3, T4), with specific parameter settings shown in Table B.2.

Table B.2

Power regulation parameter settings

Case	Initial power	Target power	Rotor-Side Converter control parameters	Grid-Side Converter control parameters
T1	-1.0	-0.9	$Kp_a = 10, Ki_a = 2$	$Kp_b = 5, Ki_b = 2$
T2	-0.9	-0.8	$Kp_a = 10, Ki_a = 2$	$Kp_b = 5, Ki_b = 2$
T3	-0.9	-1.0	$Kp_a = 10, Ki_a = 2$	$Kp_b = 5, Ki_b = 2$
T4	-0.8	-0.9	$Kp_a = 10, Ki_a = 2$	$Kp_b = 5, Ki_b = 2$

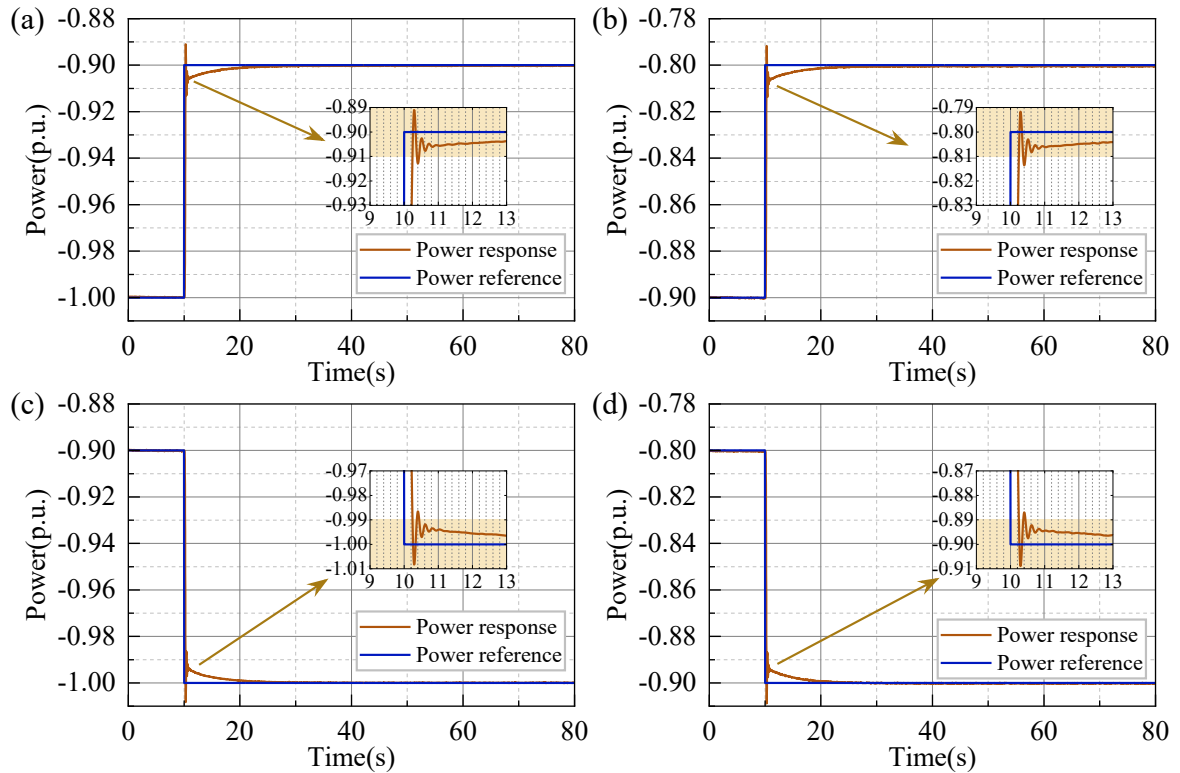


Fig. B.2. Input power reference and response time-domain plots for T1-T4

Fig. B.2(a)-(d) show the input power command and time domain response of T1-T4 respectively, where the yellow shaded area represents the $\pm 10\%$ allowable error range.

The experimental results demonstrate that under all operating conditions, the system power response can rapidly and accurately track the power command changes. Specifically: (1) Power responses under all four conditions enter the allowable error band ($\pm 10\%$) within 0.2 s, exhibiting excellent dynamic response characteristics; (2) After reaching steady state, the power response achieves near-zero error tracking relative to the command value, validating the high-precision regulation capability of the control system. Furthermore, comparing the simulation results of condition T3 with the corresponding real machine test data from Ref. [39], both show a high degree of consistency. The above analysis confirms the validity and reliability of the control system and electrical system models established in this paper.

Appendix C

The basic information of the power plant used in this paper is as follows.

Table C.1
Basic information of the power plant

Category	Parameters	Value	Units
Upstream reservoir	Normal storage water level	2332	m
	Dead water level	2300	m
Downstream reservoir	Normal storage water level	1745	m
	Dead water level	1718	m
Dynamic parameters of units under generating operations	Installed capacity	1200	MW
	Number of installed units	4	
	Rated water head	573	m
	Rated rotational speed	428.6	rpm
	Rated flow	70.3	m ³ /s
Dynamic parameters of units under pumping operations	Generator flywheel torque	6100	tm ²
	Maximum lifting height	621.9	m
	Minimum lifting height	560.5	m
	Pump input power	300	MW
	Pumping flow	50-58	m ³ /s

Appendix D

We conducted a systematic evaluation of the computational performance of the proposed framework. Taking governor parameter adjustment as an example, we measured the simulation computation time under different parameter configurations, as shown in Table D.1. Under our testing environment (6-core CPU, 16 GB RAM), the average time for a single mathematical model simulation is 16 min. The test results indicate that variations in governor parameters K_p and K_i have minimal impact on computation time, demonstrating good parameter stability of the system. Regarding scalability, the proposed evaluation framework adopts a modular design and hierarchical solution strategy, which can be conveniently extended to more FSU and VSU units while providing optimization results within an engineering-acceptable timeframe.

Table D.1
Simulation time

K_p in arithmetic sequence, K_i fixed	2.2	2.4	2.6	2.8	3.0	3.2
Simulation time	3min	3min	3min	4min	3min	4min
K_i in arithmetic sequence, K_p fixed	1.2	1.4	1.6	1.8	2.0	
Simulation time	3min	3min	4min	4min	3min	

Appendix E

To demonstrate the comparative performance of the two control strategies, we have conducted experiments examining hydraulic disturbances under both power priority control and speed priority control with different PI parameters.

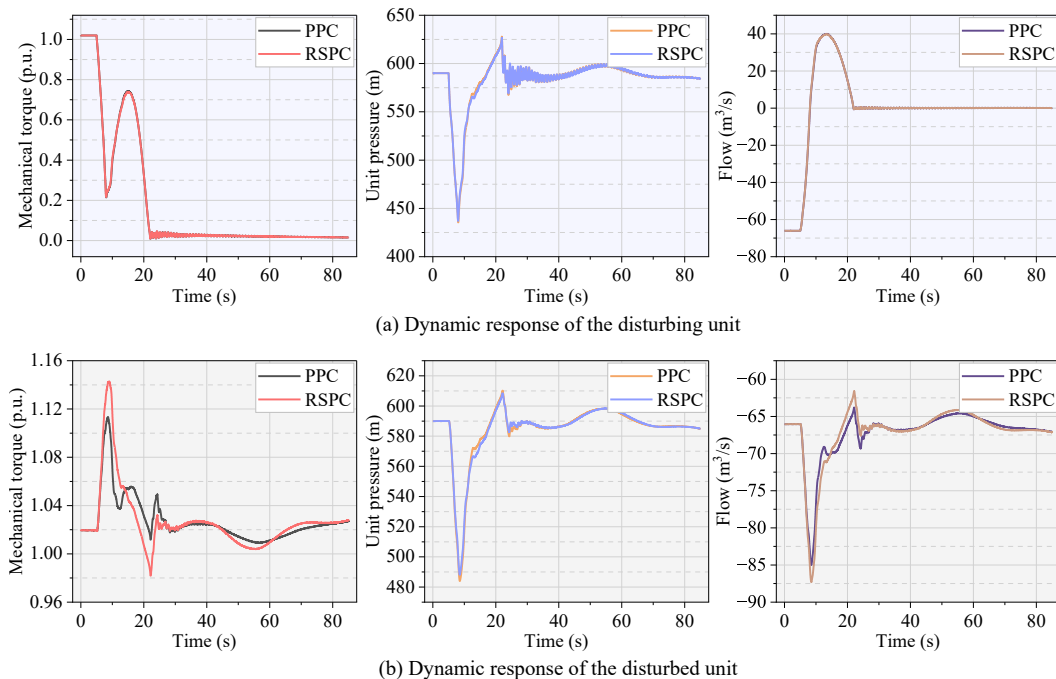


Fig. E.1. Dynamic response of PPC (Power Priority Control) and RSPC (Rotational Speed Priority Control) unit under hydraulic disturbance ($K_p = 2.7$, $K_i = 1.1$)

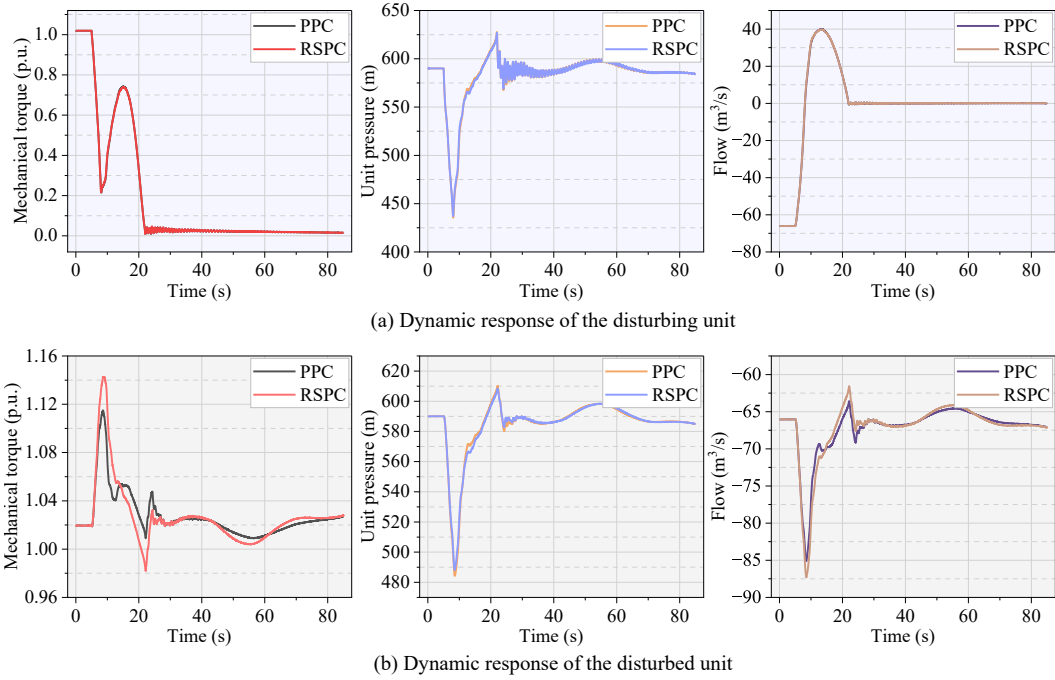


Fig. E.2. Dynamic response of PPC (Power Priority Control) and RSPC (Rotational Speed Priority Control) unit under hydraulic disturbance ($K_p = 3.5$, $K_i = 1.1$)

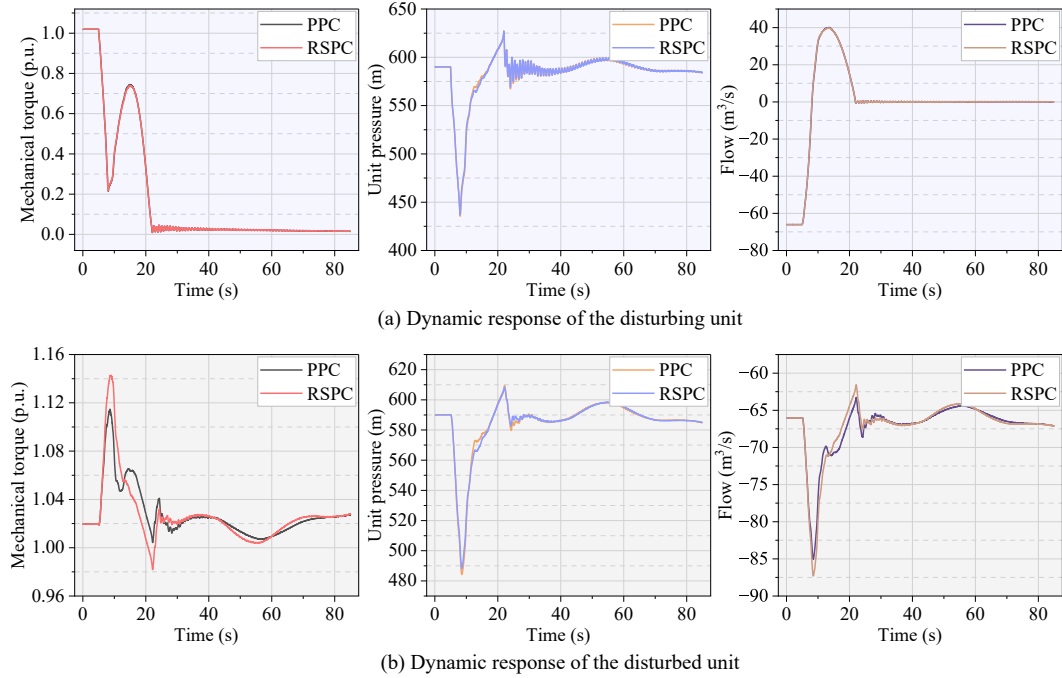


Fig. E.3. Dynamic response of PPC (Power Priority Control) and RSPC (Rotational Speed Priority Control) unit under hydraulic disturbance ($K_p = 2.7$, $K_i = 1.7$)

Figures E.1, E.2, and E.3 show the dynamic responses of PPC and RSPC under different K_p and K_i parameter combinations. The results indicate that PPC exhibits larger pressure fluctuations in the unit compared to RSPC. However, the mechanical torque fluctuations of the disturbed unit under PPC are smaller than those under RSPC. This phenomenon is consistently observed across all K_p and K_i combinations, indicating that PPC is more favorable for mechanical power regulation. Considering that RSPC demonstrates faster speed response, both control strategies have their respective advantages and disadvantages. Based on the literature review and engineering practice, the control mode selection for variable-speed units follows these general principles:

For generating mode, the control strategy depends on the operational objective. When tracking power dispatch commands or participating in automatic generation control (AGC), power priority control is typically employed. For frequency regulation or islanded operation scenarios, speed priority control is necessary to maintain system stability.

For pump mode, speed priority control (or fast speed control) is a commonly adopted strategy. As described in Ref. [40], when this strategy is employed, the unit speed is directly regulated by the converter, while the governor is responsible for adjusting the guide vane opening. The primary

reasons for adopting this approach are: (1) when the guide vane opening $GVO > 4^\circ$, the effect of guide vane adjustment on power regulation is very limited; (2) due to the cubic relationship between pump power and speed ($P \propto n^3$), variable-speed units operating at different rotational speeds possess a wider power regulation range in pump mode compared to fixed-speed units. On the other hand, power priority control is also viable; Reference [41] conducted a comparative study of power priority control and speed priority control strategies for doubly-fed variable-speed pumped storage units in pump mode, concluding that power priority control achieves better performance in power regulation. Finally, if the optimal guide vane opening has been determined, it is also feasible to control the speed using a PI governor while the electromagnetic power is controlled by the converter, which is the approach adopted in this study [42].

Data availability

Data will be made available on request.

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